22050 Signals and linear systems in continuous time

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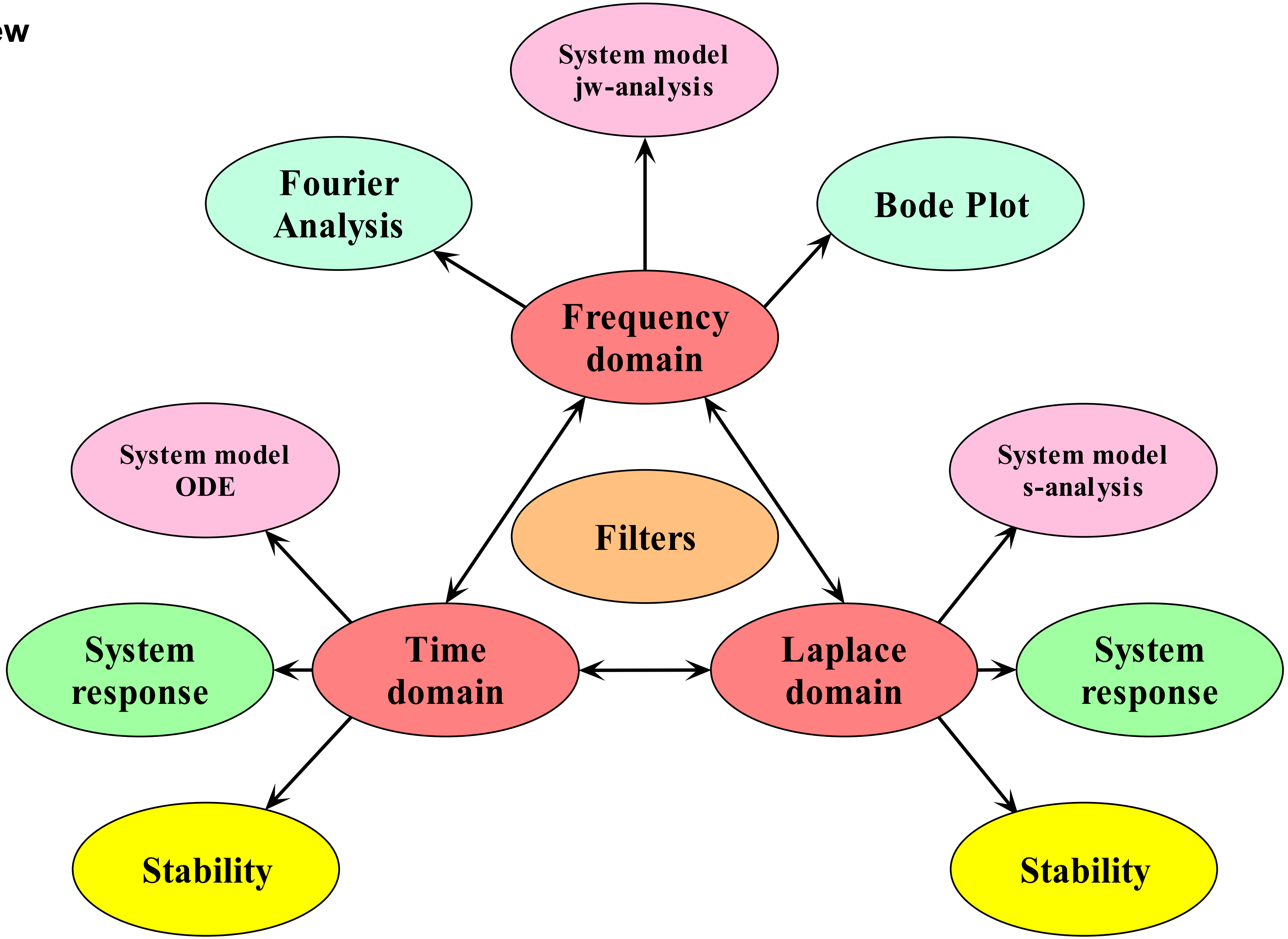
L03

Convolution
Zero-state response
Classical solutions
Stability

Course overview

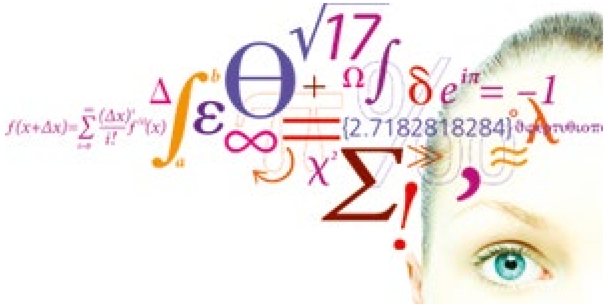
- System
  - Model
  - Response
  - Performance

- Domains
  - Time
  - Frequency
  - s-domain



DTU

Break 10 min



- Review video
  - Decomposition property
  - Zero input response
  - Unit impulse response
- Zero-state response
  - Superposition
  - Convolution integral
- Classical method
- Stability
- Problems

Review

Video 1

After lectures 1 - 3 you should be able to

After lectures 1 - 3 you should be able to

- Classify signals and systems
- Perform operations on signals
- Calculate zero-input responses
- Define the impulse response $h(t)$
- Define initial values and solve for impulse responses
- Calculate convolution integrals for zero-state responses
- Use KiCad/Spice for simulation of electrical circuits
- Use Maple or equivalent tool to perform calculations on signals and linear systems and plot signals

Properties of linear systems

Unscaled input:

$$\ddot{y}_1(t) + a_1\dot{y}_1(t) + a_0 y_1(t) = x(t)$$

$$h_1(t) = y_1(t)u(t)$$

If we differentiate on the rhs, we must differentiate on the lhs:

$$\dddot{y}_1(t) + a_1\ddot{y}_1(t) + a_0\dot{y}_1(t) = \dot{x}(t)$$

Defining new unknown:

$$y_2(t) \stackrel{\text{def}}{=} \dot{y}_1(t)$$

$$\ddot{y}_2(t) + a_1\dot{y}_2(t) + a_0 y_2(t) = \dot{x}(t)$$

$$h_2(t) = y_2(t)u(t) = \dot{y}_1(t)u(t)$$

$$b_1\ddot{y}_2(t) + a_1 b_1\dot{y}_2(t) + a_0 b_1 y_2(t) = b_1\dot{x}(t)$$

$$y_3(t) \stackrel{\text{def}}{=} b_1 y_2(t)$$

$$\ddot{y}_3(t) + a_1\dot{y}_3(t) + a_0 y_3(t) = b_1\dot{x}(t)$$

$$\begin{aligned} h_3(t) &= y_3(t)u(t) \\ &= b_1 y_2 u(t) \\ &= b_1\dot{y}_1(t)u(t) \end{aligned}$$

$$\ddot{y}_3(t) + a_1\dot{y}_3(t) + a_0 y_3(t) = P(D)x(t)$$

$$h_3(t) = P(D)\big(y_1(t)u(t)\big)$$

Response to sudden change on input

Here the input is an impulse function.

At t = 0, acceleration becomes an impulse, velocity a step function and position becomes a ramp function.

$$\ddot{y} + a_1\dot{y} + a_0 y = \delta(t), \qquad \dot{y}(0_-) = 0, y(0_-) = 0$$

$$@\ t = 0: \qquad \ddot{y} \propto \delta(t) \qquad \dot{y} \propto u(t) \qquad y \propto r(t)$$

Here the input is the derivative of the impulse function.

At t = 0, acceleration becomes the derivative of the impulse, velocity an impulse function and position becomes a step function.

$$\ddot{y} + a_1\dot{y} + a_0 y = \dot{\delta}(t), \qquad \dot{y}(0_-) = 0, y(0_-) = 0$$

$$@\ t = 0: \qquad \ddot{y} \propto \dot{\delta}(t) \qquad \dot{y} \propto \delta(t) \qquad y \propto u(t)$$

Here the input is the 2nd derivative of the impulse function.

At t = 0, acceleration becomes the 2nd derivative of the impulse, velocity is the 1st derivative of the impulse function and position becomes an impulse function.

$$\ddot{y} + a_1\dot{y} + a_0 y = \ddot{\delta}(t), \qquad \dot{y}(0_-) = 0, y(0_-) = 0$$

$$@\ t = 0: \qquad \ddot{y} \propto \ddot{\delta}(t) \qquad \dot{y} \propto \dot{\delta}(t) \qquad y \propto \delta(t)$$

Review: Solving for impulse response

The last two examples on the previous slide show that there are systems where a sudden change of input produce a synchronous sudden change on the output. In the last case the impulse response must include an impulse:

$m = n$

$$h(t) = A_0\delta(t) + \big(P(D)y_n(t)\big)u(t)$$

$$(D^n + a_{n-1}D^{n-1} + \ldots + a_1D + a_0)h(t) = (b_nD^n + b_{n-1}D^{n-1} + \ldots + b_1D + b_0)\delta(t)$$

Inserting in the differential equation we see that the coefficient of the highest order derivative of $\delta(t)$ must be the same on both sides. Hence $A_0 = b_n$.

$$h(t) = b_n\delta(t) + \big(P(D)y_n(t)\big)u(t), m \leq n$$

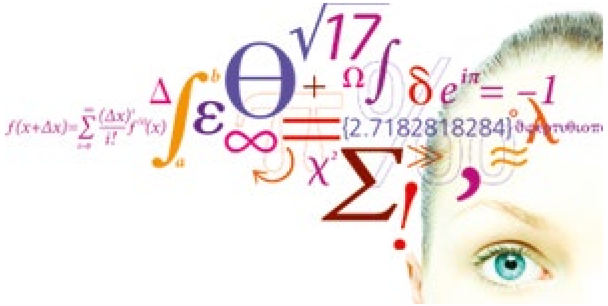
$$h(t) = P(D)\big(y_n(t)u(t)\big), m > n$$

Initial conditions:

$$y_n(0_+) = y_n^{(1)}(0_+) = \cdots = y_n^{(n-2)}(0_+) = 0$$
$$y_n^{(n-1)}(0_+) = 1$$

DTU

Break 10 min



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  - Superposition
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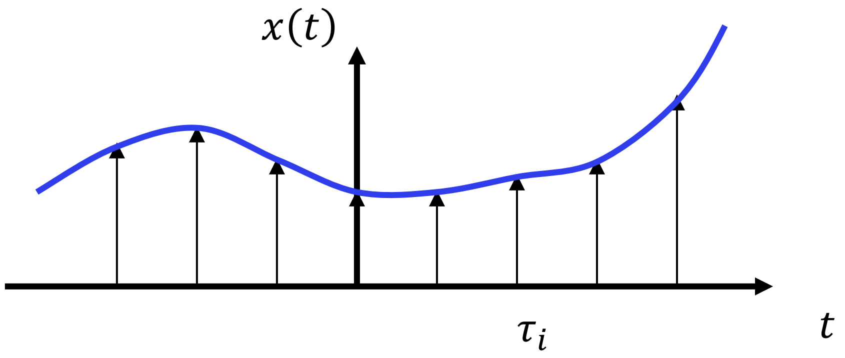
Zero-state response

Video 2

Impulse response method

If we can consider any arbitrary input signal to be a sum of the same very simple signal, then we can use the superposition principle to construct the system response.

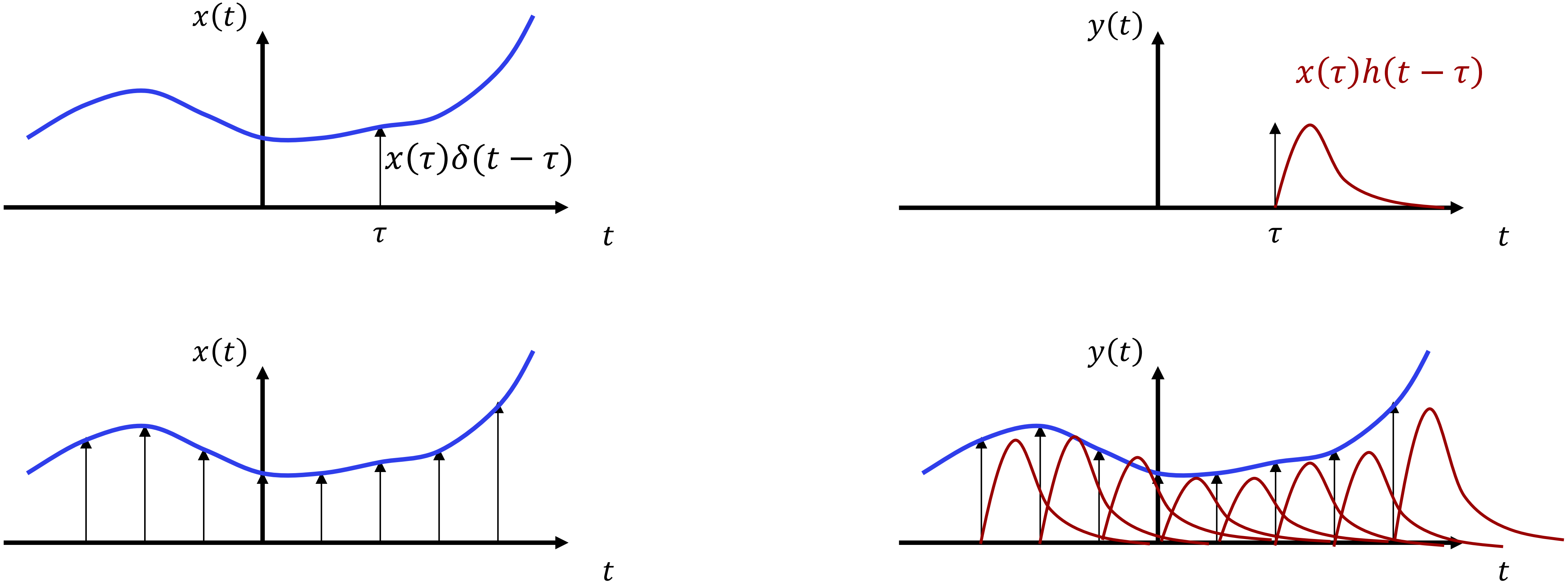
The ideal candidate is the impulse function:



In the limit as the impulse functions are placed closer and closer together, the signal $x(t)$ becomes a superposition of an infinite number of impulse functions time-shifted along the time axis.

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)d\tau$$

Superposition of impulse responses



$$\sum_{i=1}^{N} x(\tau_i)\delta(t-\tau_i)$$

$$\sum_{i=1}^{N} x(\tau_i)h(t-\tau_i)$$

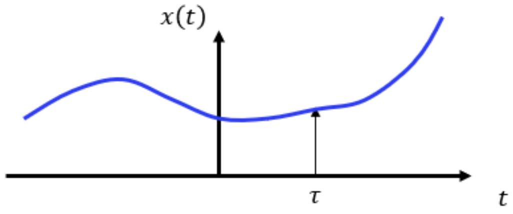
Superposition integral

Input → output

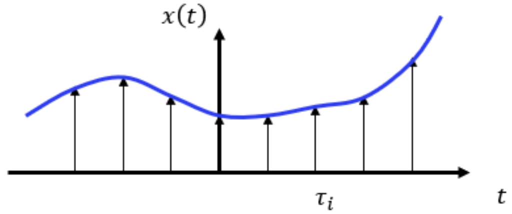
Impulse response: $\delta(t) \rightarrow h(t)$

Time invariance: $\delta(t-\tau) \rightarrow h(t-\tau)$

Homogeneity: $x(\tau)\delta(t-\tau) \rightarrow x(\tau)h(t-\tau)$



Superposition integral: $$\underbrace{\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau}_{x(t)} \rightarrow \underbrace{\int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau}_{y_{zs}(t)}$$



Convolution integral: $$y_{zs}(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$$

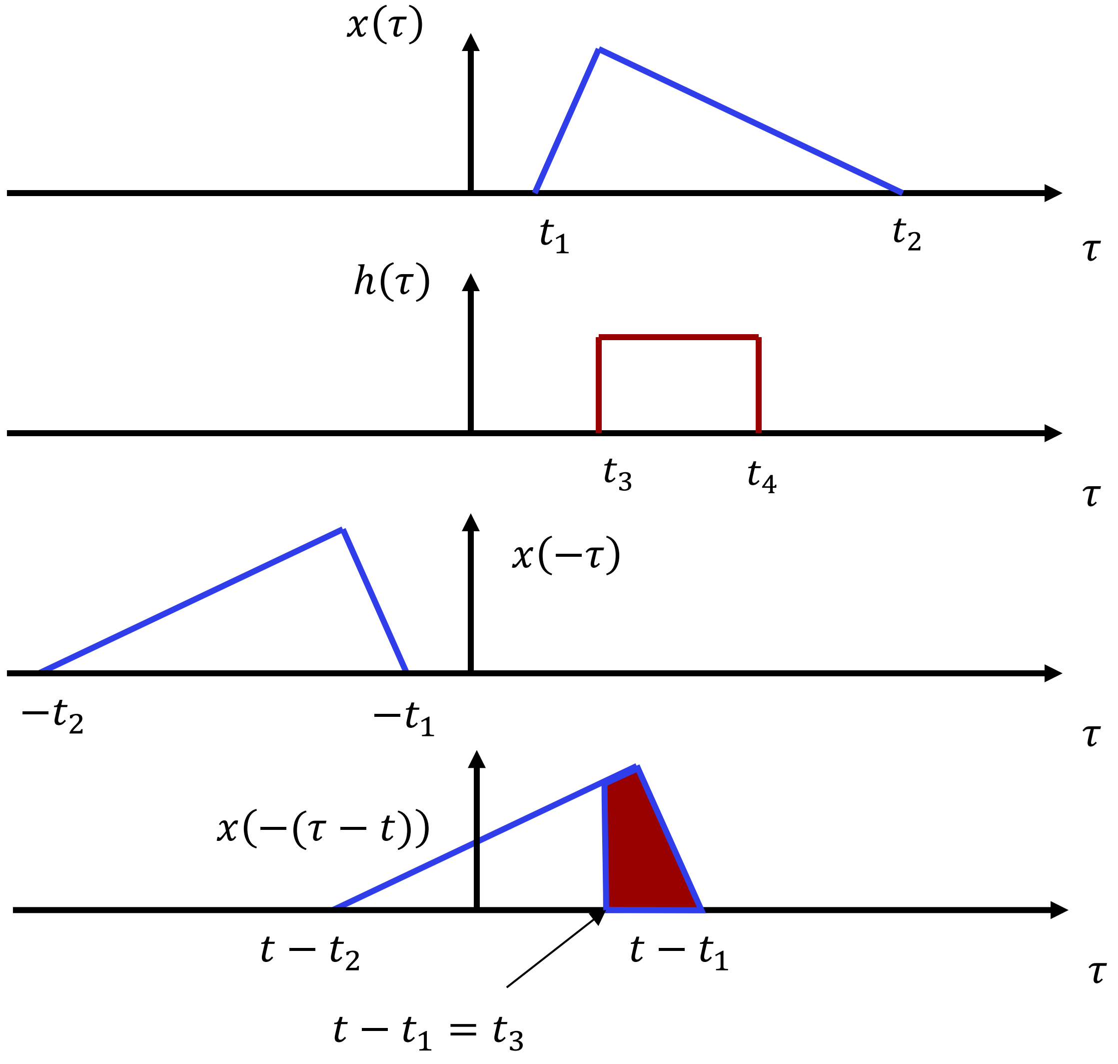
In order to calculate the zero-state response with the impulse response method, we must first determine the impulse response function $h(t)$.

Convolution

$$y_{zs}(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau) d\tau = \int_{-\infty}^{\infty} h(\tau)\, \underbrace{x\big(-(\tau - t)\big)}_{\substack{\textit{reflected and} \\ \textit{shifted toward} \\ \textit{the right}}}\, d\tau$$

Time inversion:

Time shift:



Convolution

$$y_{zs}(t) = h(t) * x(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau) d\tau$$

If we have causal *signals*:

$$x(t) = 0, \qquad t < 0_-$$

$$y_{zs}(t) = h(t) * x(t) = \int_{0_-}^{\infty} x(\tau)\, h(t-\tau) d\tau$$

If we have causal *systems*:

$$h(t) = 0, \qquad t < 0 \qquad h(t-\tau) = 0, \qquad \tau > t$$

$$y_{zs}(t) = h(t) * x(t) = \int_{-\infty}^{t} x(\tau) h(t-\tau)\, d\tau$$

If we have both:

$$y_{zs}(t) = h(t) * x(t) = \int_{0_-}^{t} x(\tau) h(t-\tau)\, d\tau$$

We will almost always have both causal signals and systems.

Convolution Properties

1. Commutative: $$x_1(t) * x_2(t) = x_2(t) * x_1(t)$$

2. Distributive: $$x_1(t) * [x_2(t) + x_3(t)] = x_1(t) * x_2(t) + x_1(t) * x_3(t)$$

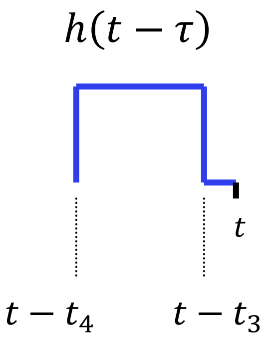
3. Associative: $$x_1(t) * [x_2(t) * x_3(t)] = [x_1(t) * x_2(t)] * x_3(t)$$

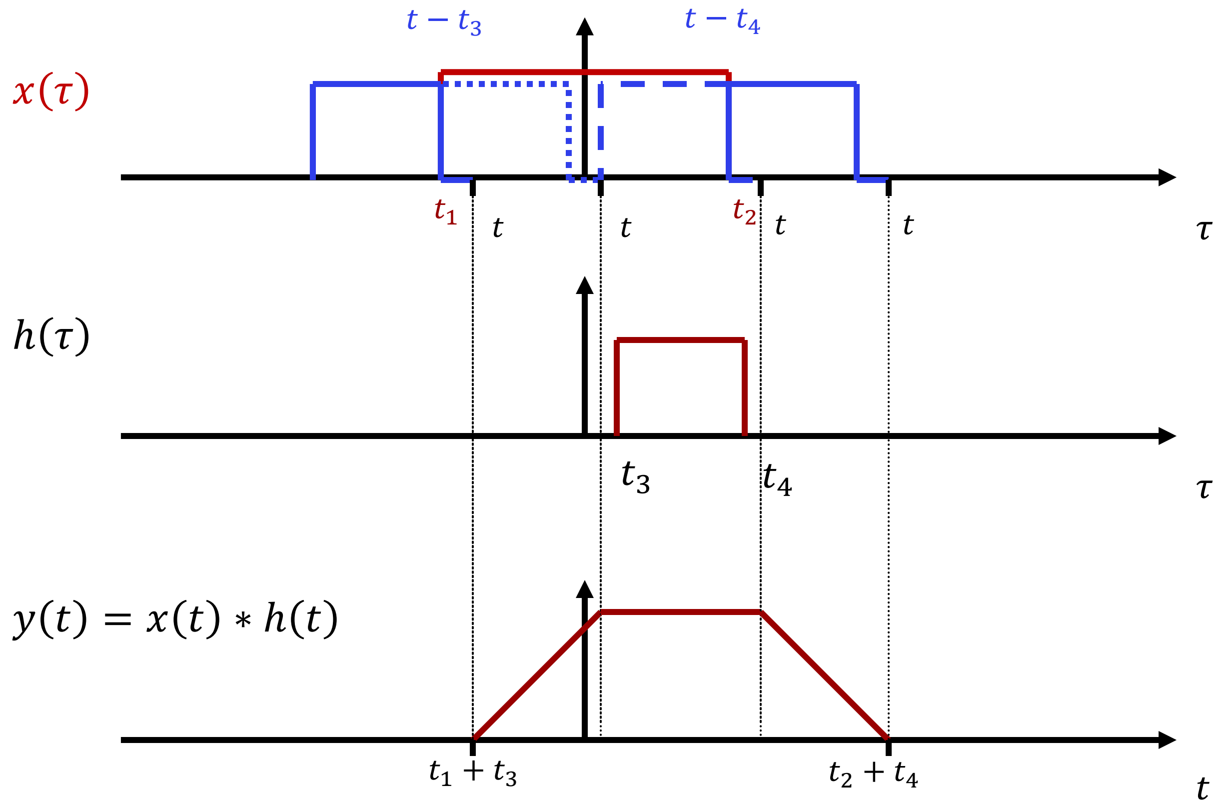
4. Time shift:
$$\begin{gathered} x_1(t) * x_2(t) = c(t) \\ x_1(t - T_1) * x_2(t) = c(t - T_1) \\ x_1(t) * x_2(t - T_2) = c(t - T_2) \\ x_1(t - T_1) * f_2(t - T_2) = c(t - T_1 - T_2) \end{gathered}$$

5. Identity: $$x(t) * \delta(t) = x(t)$$

Convolution Properties

6. Width



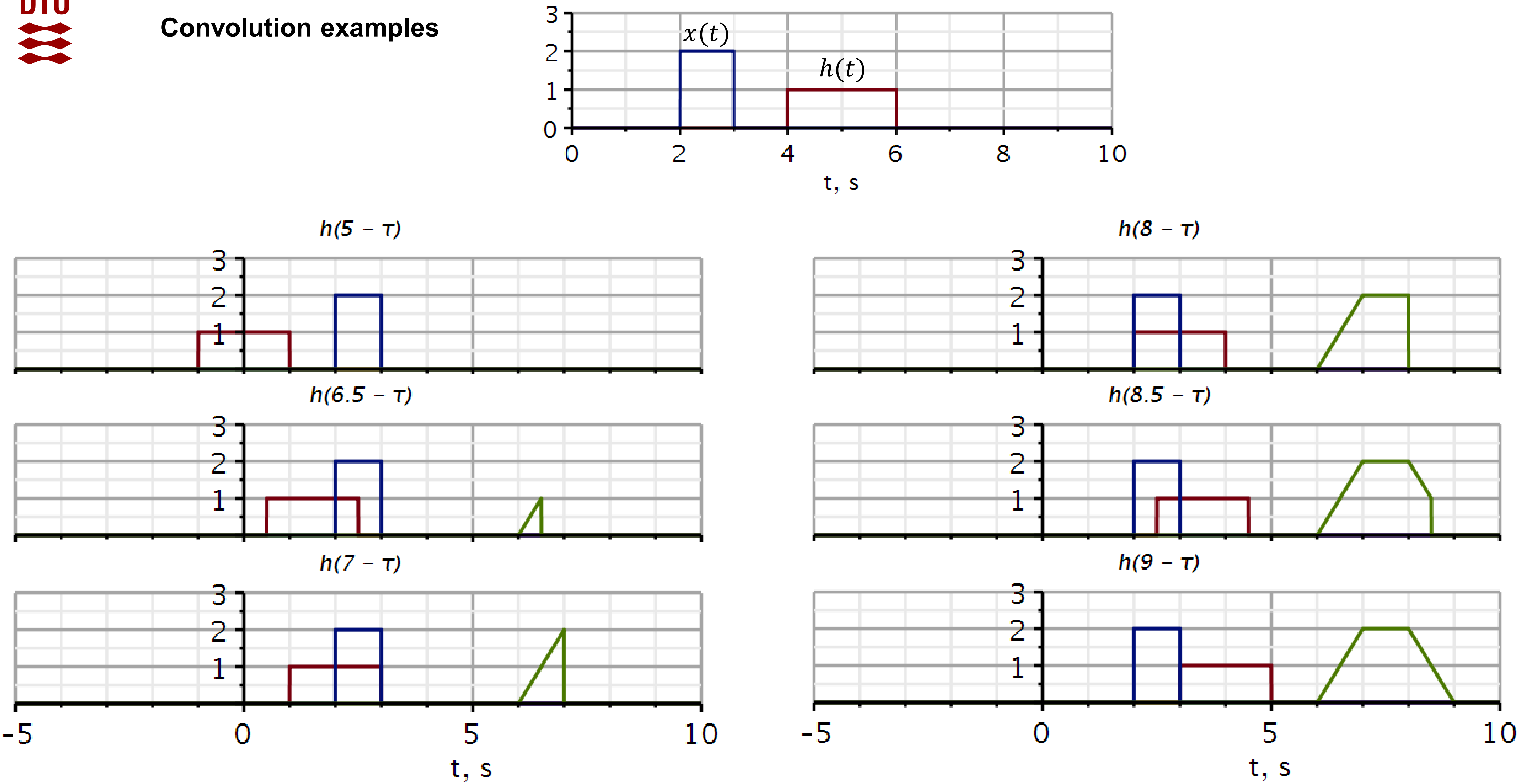


Useful check of results:

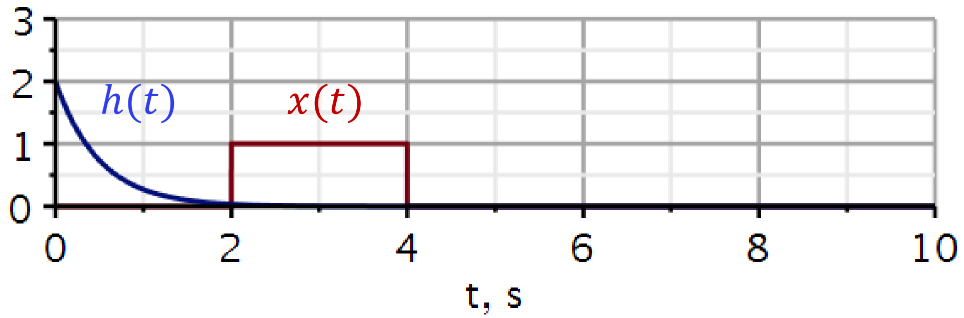
$$T_y = (t_2 + t_4) - (t_1 + t_3) = (t_2 - t_1) + (t_4 - t_3) = T_x + T_h$$

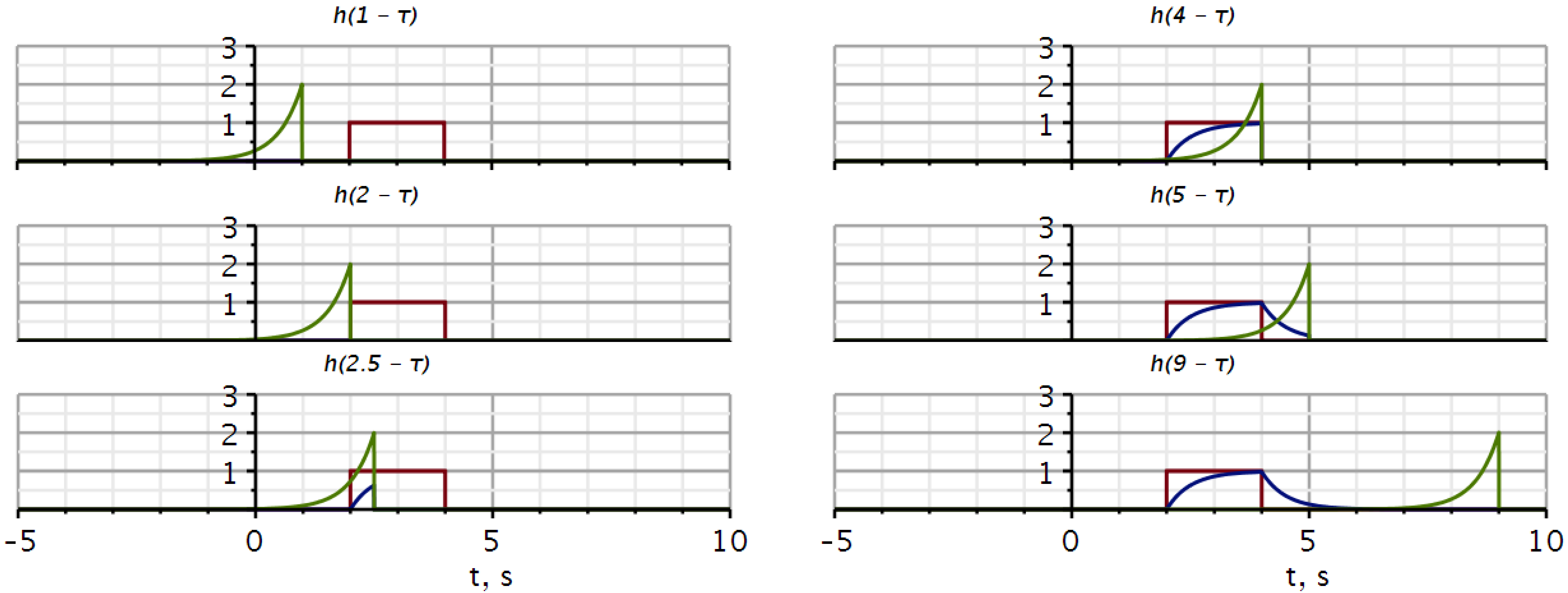
- The *starting point* of the result is the sum of the starting points of the two signals.
- The *end point* of the result is the sum of the end points of the two signals.
- The *duration* of the result is the sum of the durations of the two signals.

Convolution examples



Convolution examples





Tools in Maple

```
restart
with(plots) :
with(DynamicSystems) :
v := t→Heaviside(t) :
conv := (x, h) → t → int(x(τ) * h(t − τ), τ = 0 ..t);
```

$$conv := (x, h) \mapsto t \mapsto \int_0^t x(\tau) \cdot h(t-\tau)\, d\tau$$

Example signals

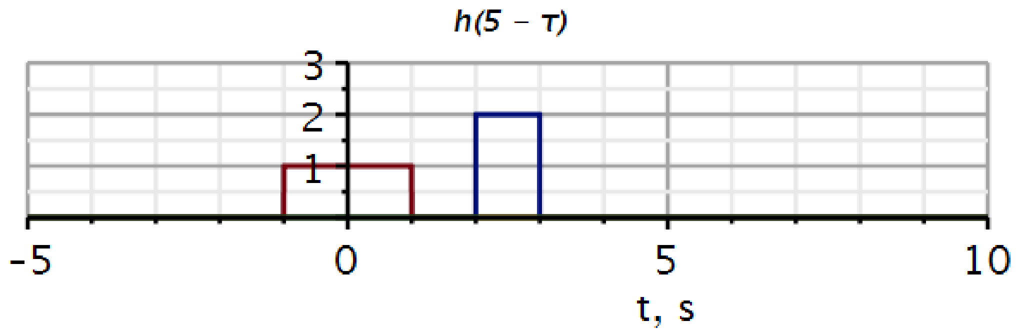
```
x1 := t→2·(v(t − 2) − v(t − 3)) :          # input signal
h1 := t→(v(t − 4) − v(t − 6)) :            # impulse response
y1 := t→conv(t→x1(t), t→h1(t))(t)          # output signal
```

$$y1 := t \mapsto conv(t \mapsto x1(t), t \mapsto h1(t))(t)$$

```
h1r := (t, T)→h1(T − t) :
    # impulse response reversed and shiftet T
y11 := (t, T)→piecewise(0 < T − t, y1(t))  # truncate output at T
```

$$y11 := (t, T) \mapsto \{\, y1(t) \quad 0 < T - t$$

```
convT := T→plot({x1(t), h1r(t, T), y11(t, T)}, t=-5 ..10, 0 ..3, thickness = 3,
    axesfont = ["Helvetica", "ROMAN", 18], axis[2] = [thickness = 2.5], axis[1]
    = [mode = linear, thickness = 2.5], labels = ["t, s" , " "], labelfont
    = ["HELVETICA", 18], gridlines, size = [600, 200], titlefont = ["Helvetica",
    "ITALIC", 14], title = cat("h(", convert(T, string), " - τ)")) :
    # a function to plot signals at T
convT(5)
```



Tools in Maple

Example signals

$h2 := t \rightarrow piecewise\left(t < 5,\ 2 \cdot e^{-2 \cdot t}\right) :$ # impulse response
$x2 := t \rightarrow (v(t-2) - v(t-4)) :$ # input signal
$y2 := t \rightarrow conv(t \rightarrow x2(t),\ t \rightarrow h2(t))(t)$ # output signal

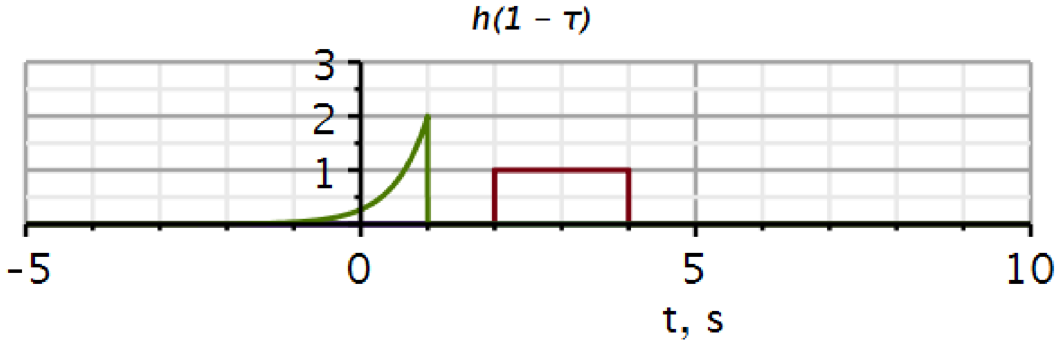
$$y2 := t \mapsto conv(t \mapsto x2(t),\ t \mapsto h2(t))(t)$$

$\boxed{h2r := (t, T) \rightarrow piecewise(0 < T - t,\ h2(T-t)) :}$
impulse response reversed and shiftet T
$y21 := (t, T) \rightarrow piecewise(0 < T - t,\ y2(t))$ # truncate output at T

$$y21 := (t, T) \mapsto \begin{cases} y2(t) & 0 < T - t \end{cases}$$

$convT2 := T \rightarrow plot(\{x2(t), h2r(t, T), y21(t, T)\}, t = -5..10, 0..3, thickness = 3,$ $axesfont = ["Helvetica", "ROMAN", 18], axis[2] = [thickness = 2.5], axis[1]$ $= [mode = linear, thickness = 2.5], labels = ["t, s", " "], labelfont$ $= ["HELVETICA", 18], gridlines, size = [600, 200], titlefont = ["Helvetica",$ $"ITALIC", 14], title = cat("h(", convert(T, string), " - \tau)")) :$
a function to plot signals at T

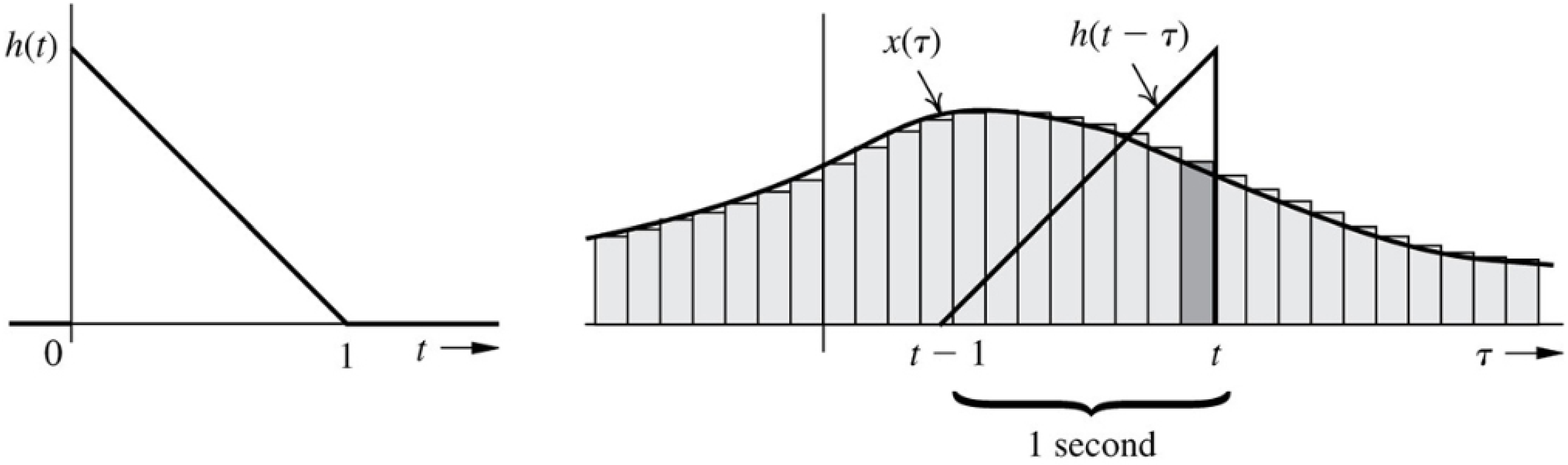
$convT2(1)$



Notice that the impulse response is truncated at $t = 5$.

The reversed impulse response is truncated at T.

Impulse response weighs previous inputs



The output of the convolution can be thought of as a weighted sum (integral) of previous inputs. The most resent input contributes with the highest weight, older inputs contributes to the output with smaller and smaller weights. This system has a finite memory span of 1s.

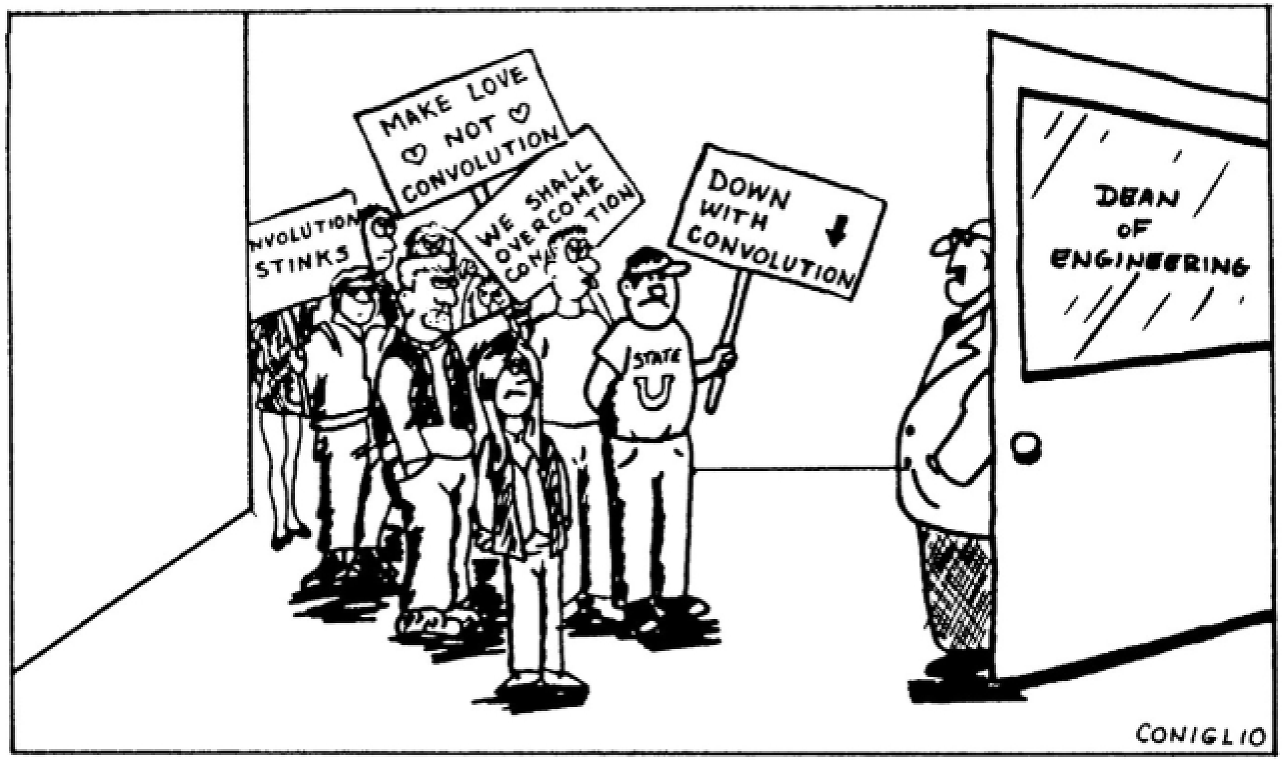
Convolution table

Do not forget the table of convolutions.

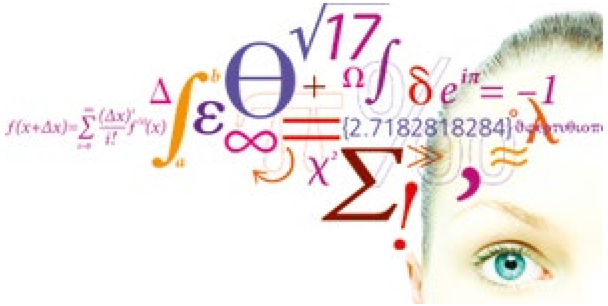
Convolution of $x(t)$ with a step function $u(t)$ is equivalent to integration of $x(t)$.

No.	$x_1(t)$	$x_2(t)$	$x_1(t)*x_2(t) = x_2(t)*x_1(t)$
1	$x(t)$	$\delta(t-T)$	$x(t-T)$
2	$e^{\lambda t}u(t)$	$u(t)$	$\frac{1-e^{\lambda t}}{-\lambda}u(t)$
3	$u(t)$	$u(t)$	$tu(t)$
4	$e^{\lambda_1 t}u(t)$	$e^{\lambda_2 t}u(t)$	$\frac{e^{\lambda_1 t}-e^{\lambda_2 t}}{\lambda_1-\lambda_2}u(t) \qquad \lambda_1 \neq \lambda_2$
5	$e^{\lambda t}u(t)$	$e^{\lambda t}u(t)$	$te^{\lambda t}u(t)$
6	$te^{\lambda t}u(t)$	$e^{\lambda t}u(t)$	$\frac{1}{2}t^2e^{\lambda t}u(t)$
7	$t^N u(t)$	$e^{\lambda t}u(t)$	$\frac{N!e^{\lambda t}}{\lambda^{N+1}}u(t) - \sum_{k=0}^{N}\frac{N!t^{N-k}}{\lambda^{k+1}(N-k)!}u(t)$
8	$t^M u(t)$	$t^N u(t)$	$\frac{M!N!}{(M+N+1)!}t^{M+N+1}u(t)$
9	$te^{\lambda_1 t}u(t)$	$e^{\lambda_2 t}u(t)$	$\frac{e^{\lambda_2 t}-e^{\lambda_1 t}+(\lambda_1-\lambda_2)te^{\lambda_1 t}}{(\lambda_1-\lambda_2)^2}u(t)$

Convolution



Break 10 min



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Classical methods

Video 3

Classical solution method for differential equations

The main difference between the classical method and the method used here is that in the classical approach, the initial conditions are defined at $t = 0_+$.

Students have been introduced to the classical method in Mat 1 and Mat 2. It is assumed that students master this method, and it will only be reviewed here with a few examples.

Classical solution techniques for ordinary differential equations (ODEs)

1. Solving for *particular* solution
2. Add *particular* solution and *homogeneous* solution
3. Solve for unknown coefficients using initial conditions at $t = 0_+$

$$Q(D)y(t) = P(D)x(t)$$
$$Q(D)\left[y_0(t) + y_\varphi(t)\right] = P(D)x(t)$$

Natural response: $Q(D)y_0(t) = 0$

Forced response: $Q(D)y_\varphi(t) = P(D)x(t)$

The right-hand side is a weighted sum of $x(t)$ and its derivatives. Thus, $y_\varphi(t)$ must be chosen so that the left-hand side can match every term on the right-hand side.

Example 1: Polynomial input

$$(D^2 + 5D + 6)y(t) = (D + 1)x(t)$$
$$x(t) = 6t^2, y(0_+) = 25/18, \dot{y}(0_+) = -2/3$$

Guess: $$y_\varphi(t) = \beta_2 t^2 + \beta_1 t + \beta_0, t > 0$$

$y_\varphi(t)$ inserted: $$2\beta_2 + 5(2\beta_2 t + \beta_1) + 6(\beta_2 t^2 + \beta_1 t + \beta_0) = 6t^2 + 12t$$

Matching coefficients for equal power terms:

$$t^2 : 6\beta_2 t^2 = 6t^2 \Rightarrow \beta_2 = 1$$
$$t^1 : 10t + 6\beta_1 t = 12t \Rightarrow \beta_1 = 1/3$$
$$t^0 : 2 + 5/3 + 6\beta_0 = 0 \Rightarrow \beta_0 = -11/18$$

Particular solution: $$y_\varphi(t) = t^2 + \frac{1}{3}t - \frac{11}{18}, t > 0$$

Example 1: Polynomial input

Homogeneous equation: $(D^2 + 5D + 6)y_0(t) = 0; y(0_+) = 25/18, \dot{y}(0_+) = -2/3$

Characteristic equation: $\lambda^2 + 5\lambda + 6 = (\lambda + 2)(\lambda + 3) = 0$

Generic solution form:

$$y_0(t) = C_1 e^{-2t} + C_2 e^{-3t}; t \geq 0$$

$$y(t) = C_1 e^{-2t} + C_2 e^{-3t} + t^2 + \frac{1}{3}t - \frac{11}{18}$$

$$\dot{y}(t) = -2C_1 e^{-2t} - 3C_2 e^{-3t} + 2t + \frac{1}{3}$$

Solving for unknown coefficients:

$$\left\{ \begin{aligned} y(0_+) &= C_1 + C_2 - \frac{11}{18} = \frac{25}{18} \\ \dot{y}(0_+) &= -2C_1 - 3C_2 + \frac{1}{3} = -\frac{2}{3} \end{aligned} \right\} \Rightarrow \left\{ \begin{aligned} C_1 &= 5 \\ C_2 &= -3 \end{aligned} \right\}$$

$$y(t) = 5e^{-2t} - 3e^{-3t} + t^2 + \frac{1}{3}t - \frac{11}{18}; t \geq 0$$

Response to exponential inputs

While the convolution method allow us to compute the response to a large number of input signals, the method does not provide clear insight into what action the system performs on the input signal.

To get some insight into this at this point in the course, we will use the classical method.

When the input signal is exponential, the differential operator polynomial produces an ordinary polynomial. Hence after all the differentiations, we end up with an algebraic equation.

$$Q(D)y(t) = P(D)x(t) \qquad D^i y(0_+) = \alpha_i, i = 0, \ldots, n-1$$

$$x(t) = e^{\gamma t}$$

$$y_\phi(t) = \beta e^{\gamma t}$$

$$De^{\gamma t} = \gamma e^{\gamma t}$$

$$D^2 e^{\gamma t} = \gamma^2 e^{\gamma t}$$

$$D^r e^{\gamma t} = \gamma^r e^{\gamma t}$$

$$Q(D)\beta e^{\gamma t} = Q(\gamma)\beta e^{\gamma t} \qquad P(D)e^{\gamma t} = P(\gamma)e^{\gamma t}$$

$$Q(D)y_\phi(t) = Q(\gamma)\beta e^{\gamma t} = P(\gamma)e^{\gamma t} = P(D)x(t)$$

$$\beta = \frac{P(\gamma)}{Q(\gamma)}$$

Response to exponential inputs

Example:

Exponential input: $x(t) = e^{\gamma\, t}$

Exponential output: $y_\phi(t) = \beta e^{\gamma t}, \qquad \beta = gain$

Differential equation:

$$\underbrace{(D^2 + a_1 D + a_0)}_{Q(D)} \beta e^{\gamma t} = \underbrace{(b_1 D + b_0)}_{P(D)} e^{\gamma t}$$

Algebraic equation:

$$\underbrace{(\gamma^2 + a_1\gamma + a_0)}_{Q(\gamma)} \beta e^{\gamma t} = \underbrace{(b_1\gamma + b_0)}_{P(\gamma)} e^{\gamma t}$$

$$\underbrace{(\gamma^2 + a_1\gamma + a_0)}_{Q(\gamma)} \beta = \underbrace{(b_1\gamma + b_0)}_{P(\gamma)}$$

We observe that the gain β is a function of γ.

$$\beta(\gamma) = \frac{(b_1\gamma + b_0)}{(\gamma^2 + a_1\gamma + a_0)} = \frac{P(\gamma)}{Q(\gamma)}$$

$$y_\phi(t) = \underbrace{\beta(\gamma)}_{gain} e^{\gamma t} = \frac{P(\gamma)}{Q(\gamma)} x(t)$$

Response to exponential inputs

We call this gain the **Transfer function** because it shows how the input is transferred to the output.

$$H(\gamma) \stackrel{\text{def}}{=} \frac{P(\gamma)}{Q(\gamma)}$$

The forced response (particular solution) is:

$$y_\phi(t) = \beta(\gamma)e^{\gamma t} = \underbrace{H(\gamma)}_{gain}\, e^{\gamma t}$$

The complete solution is the sum of the natural response and the particular solution:

$$\begin{aligned} y(t) &= y_0(t) + y_\phi(t) \\ &= \underbrace{\sum_{j=1}^{n} K_j e^{\lambda_j t}}_{\substack{natural \\ response}} + \underbrace{H(\gamma)e^{\gamma t}}_{\substack{forced \\ response}}, \qquad \gamma \neq \lambda_j \end{aligned}$$

Response to sinusoidal inputs

Let us investigate how **sinusoidal signals** of different frequencies are transferred through the system.

$$e^{\gamma\, t} = e^{j\omega t} = \cos(\omega\, t) + j\sin(\omega\, t)$$
$$e^{-\gamma\, t} = e^{-j\omega t} = \cos(\omega\, t) - j\sin(\omega\, t)$$

Adding two complex conjugated numbers yields twice the real part:

We see that the modulus of the transfer function defines the system gain as a function of frequency:

$$y_\phi(t) = H(\gamma)e^{\gamma\, t} \qquad H(\gamma) \stackrel{\text{def}}{=} \frac{P(\gamma)}{Q(\gamma)}$$

$$y_{\phi,1}(t) = H(j\omega)e^{j\omega t} \qquad y_{\phi,2}(t) = H(-j\omega)e^{-j\omega t}$$

$$y_\phi(t) = \frac{1}{2}H(j\omega)e^{j\omega t} + \frac{1}{2}H(-j\omega)e^{-j\omega t} = Re\{H(j\omega)e^{j\omega t}\}$$

$$y_\phi(t) = Re\{|H(j\omega)|e^{j\angle H(j\omega)}e^{j\omega t}\}$$

$$y_\phi(t) = Re\{|H(j\omega)|e^{\,j(\omega t + \angle H(j\omega))}\}$$

$$y_\phi(t) = |H(j\omega)|Re\{e^{\,j(\omega t + \angle H(j\omega))}\}$$

$$y_\phi(t) = \underbrace{|H(j\omega)|}_{gain}\cos\left(\omega t + \underbrace{\angle H(j\omega)}_{phase}\right)$$

Response to sinusoidal inputs

$$H(\gamma) \stackrel{\text{def}}{=} \frac{P(\gamma)}{Q(\gamma)} \qquad H(j\omega) \stackrel{\text{def}}{=} \frac{P(j\omega)}{Q(j\omega)}$$

Transfer functions

Two real roots: $\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = 2x(t)$

$$H(j\omega) \stackrel{\text{def}}{=} \frac{P(j\omega)}{Q(j\omega)} = \frac{2}{(j\omega)^2 + 3(j\omega) + 2}$$

$$|H(j\omega)| = \frac{2}{\sqrt{(2-\omega^2)^2 + (3\,\omega)^2}}$$

A double root: $\ddot{y}(t) + 6\dot{y}(t) + 9y(t) = 10\ddot{x}(t)$

$$H(j\omega) \stackrel{\text{def}}{=} \frac{P(j\omega)}{Q(j\omega)} = \frac{10(j\omega)^2}{(j\omega)^2 + 6(j\omega) + 9}$$

$$|H(j\omega)| = \frac{10\omega^2}{\sqrt{(9-\omega^2)^2 + (6\,\omega)^2}}$$

Complex conjugated roots: $\ddot{y}(t) + 4\dot{y}(t) + 13y(t) = 4\dot{x}(t)$

$$H(j\omega) \stackrel{\text{def}}{=} \frac{P(j\omega)}{Q(j\omega)} = \frac{4(j\omega)}{(j\omega)^2 + 4(j\omega) + 13}$$

$$|H(j\omega)| = \frac{4\omega}{\sqrt{(13-\omega^2)^2 + (4\,\omega)^2}}$$

Response to sinusoidal inputs

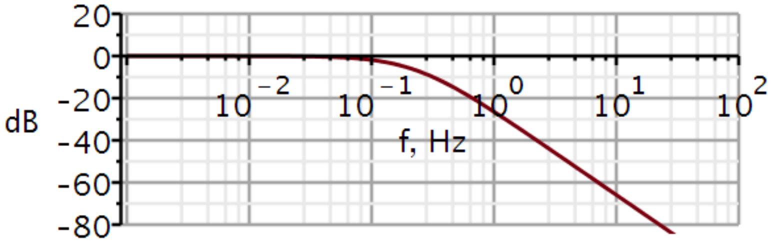
$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = 2x(t)$$

$$H(j\omega) = \frac{2}{(j\omega)^2 + 3(j\omega) + 2}$$

$$|H(j\omega)| = \frac{2}{\sqrt{(2-\omega^2)^2 + (3\,\omega)^2}}$$

$$|H(j\omega)|_{\omega\to 0} = \frac{2}{2} = 1$$

$$|H(j\omega)|_{\omega\to \infty} = \frac{2}{\infty} = 0$$



Filter? Low pass filter

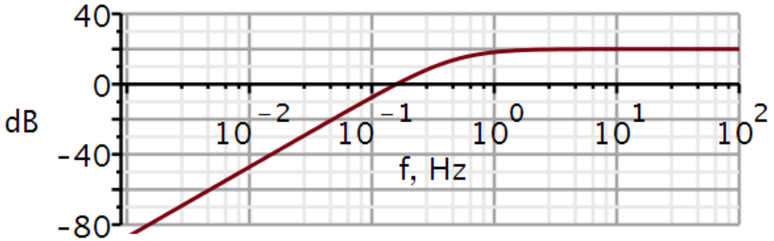
$$\ddot{y}(t) + 6\dot{y}(t) + 9y(t) = 10\ddot{x}(t)$$

$$H(j\omega) = \frac{10(j\omega)^2}{(j\omega)^2 + 6(j\omega) + 9}$$

$$|H(j\omega)| = \frac{10\omega^2}{\sqrt{(9-\omega^2)^2 + (6\,\omega)^2}}$$

$$|H(j\omega)|_{\omega\to 0} = \frac{0}{9} = 0$$

$$|H(j\omega)|_{\omega\to \infty} = \frac{10\omega^2}{\omega^2} = 10$$



Filter? High pass filter

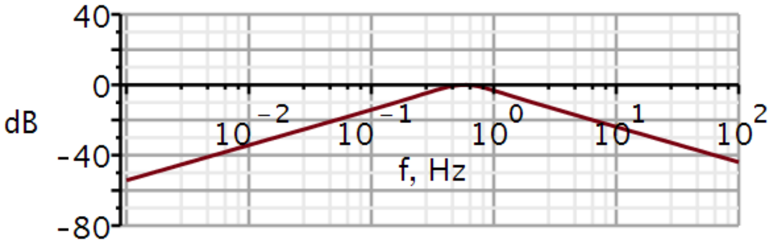
$$\ddot{y}(t) + 4\dot{y}(t) + 13y(t) = 4\dot{x}(t)$$

$$H(j\omega) = \frac{4(j\omega)}{(j\omega)^2 + 4(j\omega) + 13}$$

$$|H(j\omega)| = \frac{4\omega}{\sqrt{(13-\omega^2)^2 + (4\,\omega)^2}}$$

$$|H(j\omega)|_{\omega\to 0} = \frac{0}{13} = 0$$

$$|H(j\omega)|_{\omega\to \infty} = \frac{4}{\omega} = 0$$



Filter? Band pass filter

Maple tools for plotting transfer function

restart
with(*plots*) :
$dB := \omega \rightarrow 20 \cdot \log 10(|H(\omega)|)$: # converts to decibel
$angle := \omega \rightarrow \text{argument}(H(\omega)) \cdot \frac{180}{\pi}$: # *converts from radians to degrees*

Transfer function

$$H := \omega \rightarrow \frac{b2 \cdot (j \cdot \omega)^2 + b1 \cdot (j \cdot \omega) + b0}{(j \cdot \omega)^2 + a1 \cdot (j \cdot \omega) + a0} :$$ # *2nd order transfer function*

Example system

$b2 := 0$:; $b1 := 0$:; $b0 := 2$: # *numerator coefficients*
$a1 := 3$:; $a0 := 2$: # *denominator coefficients*

Plotting solution

semilogplot($dB(2 \cdot \pi \cdot f)$, f = 0.001 ..1E2, -80 ..20, *thickness* = 3, *font* = ["Helvetica", "roman", 18], *axis*[2] = [*thickness* = 2.5], *axis*[1] = [*thickness* = 2.5], *labels* = ["f, Hz" , "dB"], *labelfont* = ["HELVETICA", 18], *numpoints* = 100, *gridlines*, *size* = [600, 200])

semilogplot($angle(2 \cdot \pi \cdot f)$, f = 0.001 ...1E3, -200 ..200, *thickness* = 3, *font* = [*Helvetica*, *roman*, 18], *axis*[2] = [*thickness* = 2.5], *axis*[1] = [*thickness* = 2.5], *labels* = ["f, Hz" , " "], *labelfont* = ["HELVETICA", 18], *numpoints* = 100, *gridlines*, *size* = [600, 200])

Calculating systems output using convolution

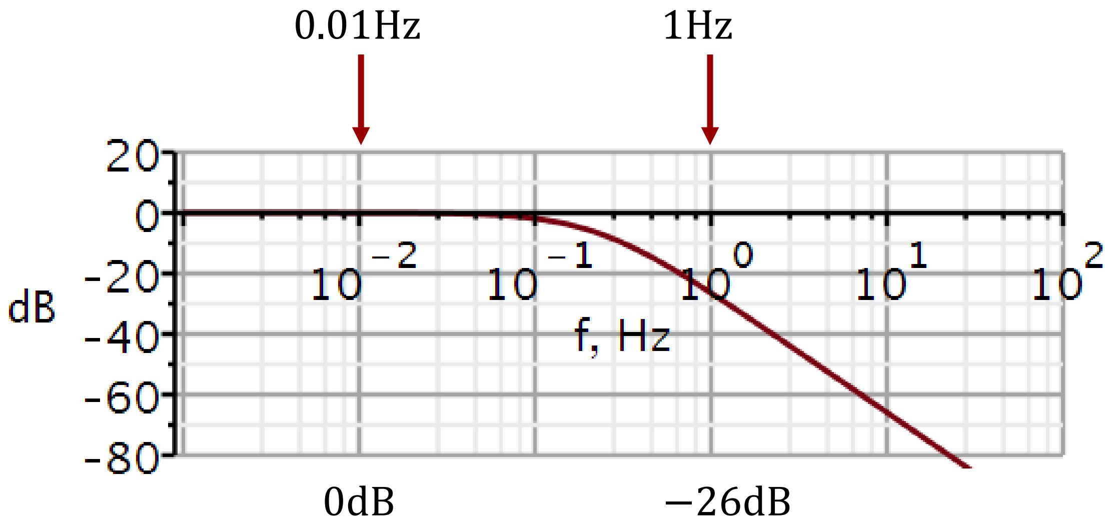
$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = 2x(t)$$

$x1 := t \rightarrow \cos(2 \cdot \pi \cdot 0.01 \cdot t) + \cos(2 \cdot \pi \cdot t)$: # *input signal*

$h1 := t \rightarrow 2 \cdot (e^{-t} - e^{-2 \cdot t})$: # *impulse response*

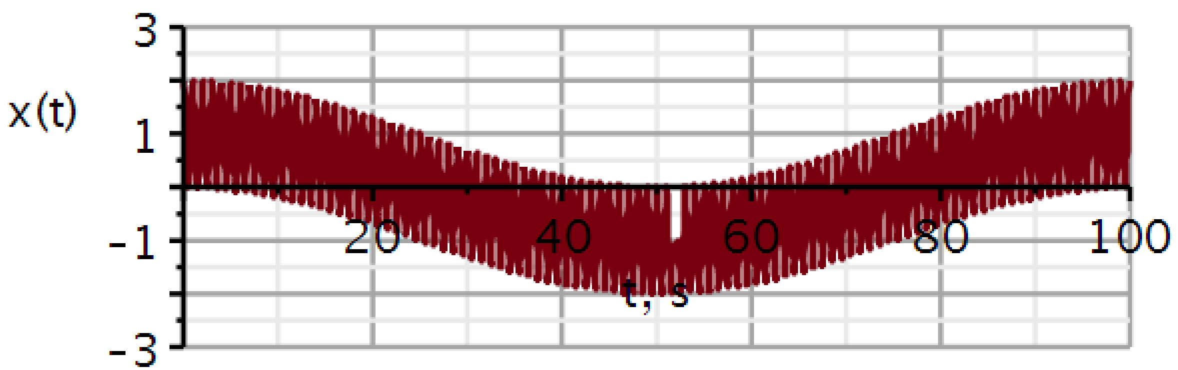
$y1 := t \rightarrow conv(t \rightarrow x1(t), t \rightarrow h1(t))(t)$ # *output signal*

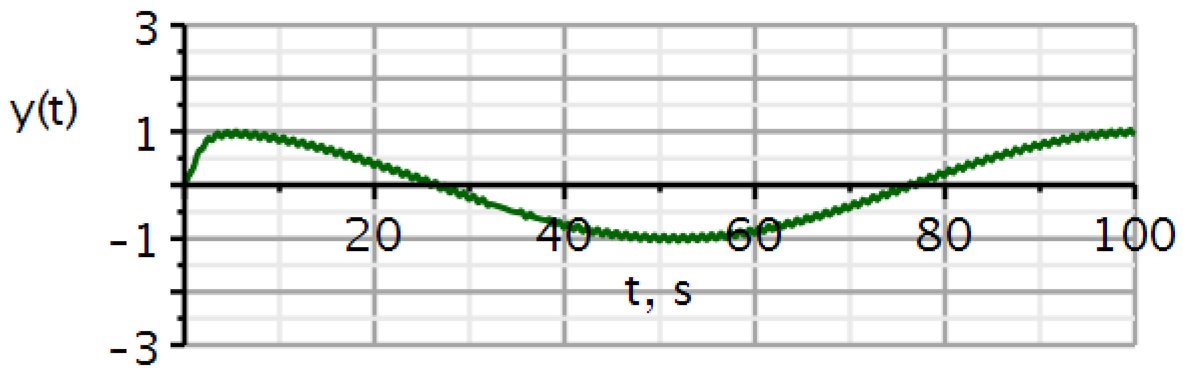
$$y1 := t \mapsto conv(t \mapsto x1(t), t \mapsto h1(t))(t)$$



$$-26\text{dB} = -20\text{dB} - 6\text{dB} \leftrightarrow 10^{-1} \times 2^{-1} = 20^{-1}$$

Input: 0.01 Hz + 1 Hz





1 Hz component removed (almost)

0.01 Hz component passed through

Calculating systems output using convolution

$$\ddot{y}(t) + 6\dot{y}(t) + 9y(t) = 10\ddot{x}(t) \qquad h(t) = 10\delta(t) + 10\frac{d^2}{dt^2}\left(t\,e^{-3t}\right)$$

$x1 := t \rightarrow \cos(2\cdot\pi\cdot 0.01\cdot t) + \cos(2\cdot\pi\cdot t)$: *# input signal*

$10\cdot\frac{d^2}{dt^2}\left(t\cdot e^{-3\cdot t}\right)$

$$-60\,e^{-3t} + 90\,t\,e^{-3t}$$

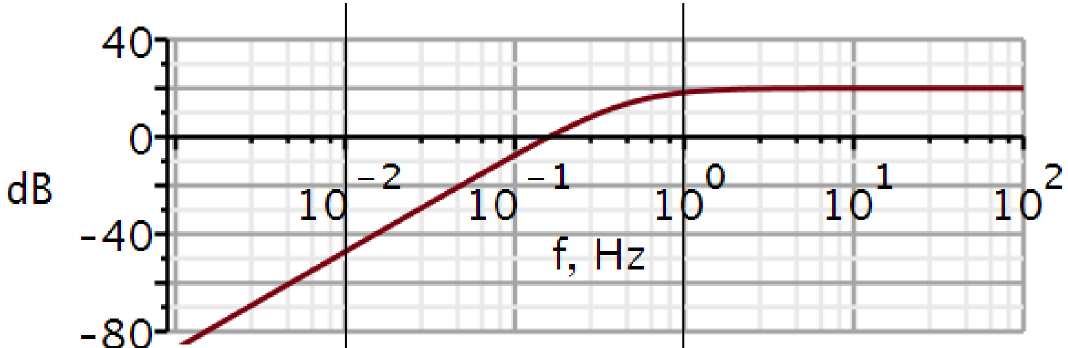
$h2 := t \rightarrow -60\,e^{-3t} + 90\,t\,e^{-3t}$: *# impulse response*

$y1 := t \rightarrow conv(t \rightarrow x1(t), t \rightarrow h2(t))(t)$ *# output signal*

$$y1 := t \mapsto conv(t \mapsto x1(t), t \mapsto h2(t))(t)$$

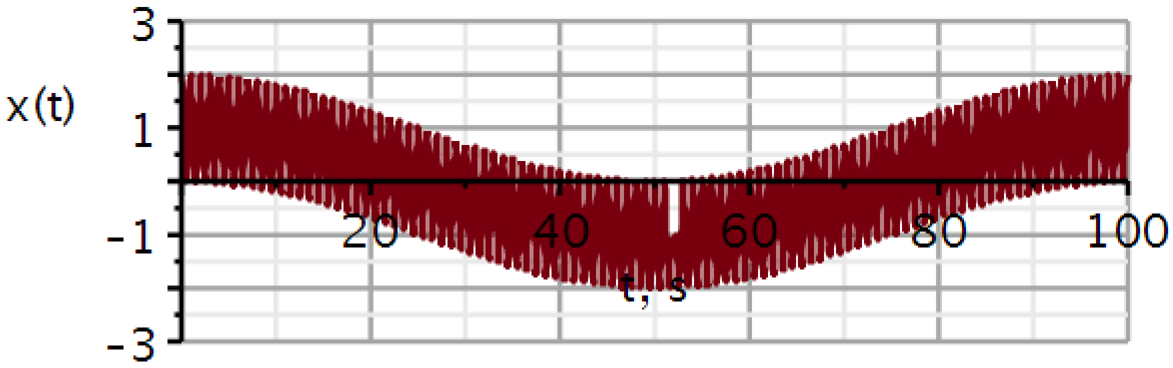
$y2 := t \rightarrow y1(t) + 10\cdot x1(t)$:

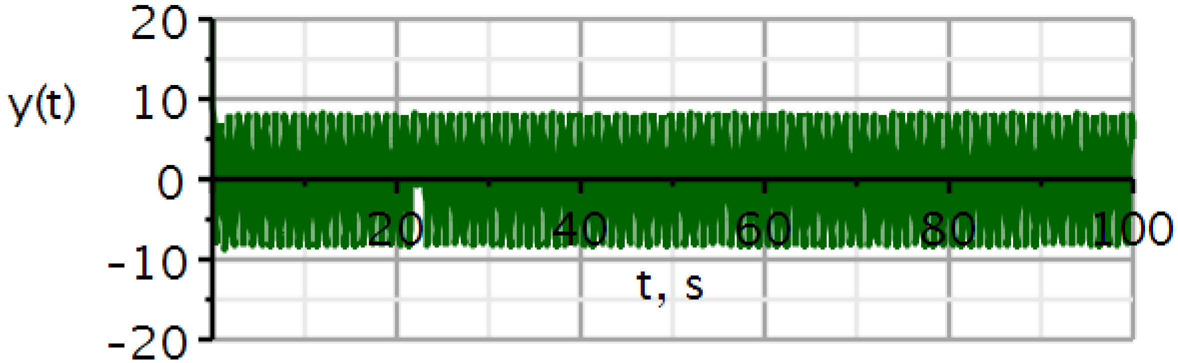
?



$-46\text{dB} = 200^{-1}$ $\qquad$ $20\text{dB} = 10$

Input: 0.01 Hz + 1 Hz





0.01 Hz component removed

1 Hz component passed through

Example 2: Natural mode input

$$(D^2 + 5D + 6)y(t) = (D + 1)x(t)$$
$$x(t) = 5e^{-2t}, y(0_+) = 1, \dot{y}(0_+) = -1$$

This is an exponential input. Let us try and use the transfer function to obtain the forced response $y_\varphi(t)$:

$$H(\gamma) = \frac{P(\gamma)}{Q(\gamma)} = \frac{\gamma + 1}{\gamma^2 + 5\gamma + 6} = \frac{\gamma + 1}{(\gamma + 2)(\gamma + 3)}$$

By finding the roots of the characteristic equation, or as here the poles of H, we see that we are heading for trouble.

$$y_\varphi(t) = H(-2)x(t) = \left.\frac{\gamma + 1}{(\gamma + 2)(\gamma + 3)}\right|_{\gamma=-2} 5e^{-2t}$$
$$= 5\frac{-2 + 1}{(-2 + 2)(-2 + 3)} e^{-2t} = \frac{-5}{0} e^{-2t}$$

If the input is a scaled copy of one of the natural modes, we must use a different approach.

Example 2: Natural mode input

Particular solution assuming double root: $y_\varphi(t) = \beta t e^{\gamma t}; t > 0$

$$D(\beta t e^{\gamma t}) = \beta e^{\gamma t} + \beta t \gamma e^{\gamma t}$$

$$D^2(\beta t e^{\gamma t}) = 2\beta\gamma e^{\gamma t} + \beta t \gamma^2 e^{\gamma t}$$

$$(D^2 + 5D + 6)(\beta t e^{\gamma t}) = (D + 1)(5e^{\gamma t})$$

$$2\beta\gamma e^{\gamma t} + \beta t \gamma^2 e^{\gamma t} + 5 \cdot (\beta e^{\gamma t} + \beta t \gamma e^{\gamma t}) + 6 \cdot \beta t e^{\gamma t} = 5 \cdot (\gamma e^{\gamma t} + e^{\gamma t})$$

$$2\beta\gamma + \beta t \gamma^2 + 5 \cdot (\beta + \beta t \gamma) + 6 \cdot \beta t = 5 \cdot (\gamma + 1)$$

Matching left-side and right-side terms with t:

$$\beta t \gamma^2 + 5 \cdot (\beta t \gamma) + 6 \cdot \beta t = 0$$

$$\gamma = -2 \qquad \beta t 4 - 5 \cdot (\beta t 2) + 6 \cdot \beta t = (10 - 10)\beta t = 0$$

These terms turn out to be independent of β

Matching left-side and right-side terms without t:

$$2\beta(-2) + 5 \cdot (\beta) = 5 \cdot (-2 + 1) \Rightarrow \beta = -\frac{5}{5-4} = -5$$

$$y_\varphi(t) = -5te^{-2t}; t > 0$$

Example 2: Natural mode input

Homogeneous equation:

$$(D^2 + 5D + 6)y_n(t) = 0$$

$e^{\lambda t}$

$$Q(\lambda) = \lambda^2 + 5\lambda + 6 = 0 \Rightarrow \lambda_1 = -2, \lambda_2 = -3$$

$$y_n(t) = K_1 e^{-2t} + K_2 e^{-3t}$$

$$y(t) = K_1 e^{-2t} + K_2 e^{-3t} - 5te^{-2t}$$

$$\dot{y}(t) = -2K_1 e^{-2t} - 3K_2 e^{-3t} - 5e^{-2t} + 10te^{-2t}$$

$$y(0_+) = 1, \dot{y}(0_+) = -1$$

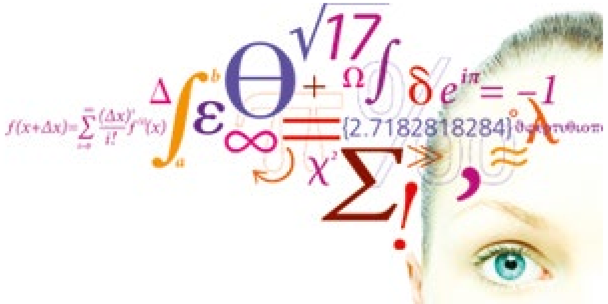
$$y(0_+) = K_1 + K_2 = 1$$

$$\dot{y}(0_+) = -2K_1 - 3K_2 - 5 = -1$$

$$\Rightarrow K_1 = 7, K_2 = -6$$

$$y(t) = \left((7 - 5t)e^{-2t} - 6e^{-3t}\right)u(t)$$

Break 10 min



- Review
  - Decomposition property
  - Zero input response
  - Unit impulse response
- Zero-state response
  - Superposition
  - Convolution integral
- Classical method
- Stability
- Problems

Stability

$$Q(D)y(t) = P(D)x(t)$$

Characteristic equation: $Q(\lambda) = 0$

For unrepeated roots:

Natural response: $y_0(t) = \sum_{j=1}^{n} C_j e^{\lambda_j t}, t \geq 0$

Asymptotically stable: $Re\{\lambda_j\} < 0,$ exponentially decreasing response

Unstable system: $Re\{\lambda_j\} > 0,$ exponentially increasing response

Marginally stable: $Re\{\lambda_j\} = 0,$ oscillating with constant amplitude

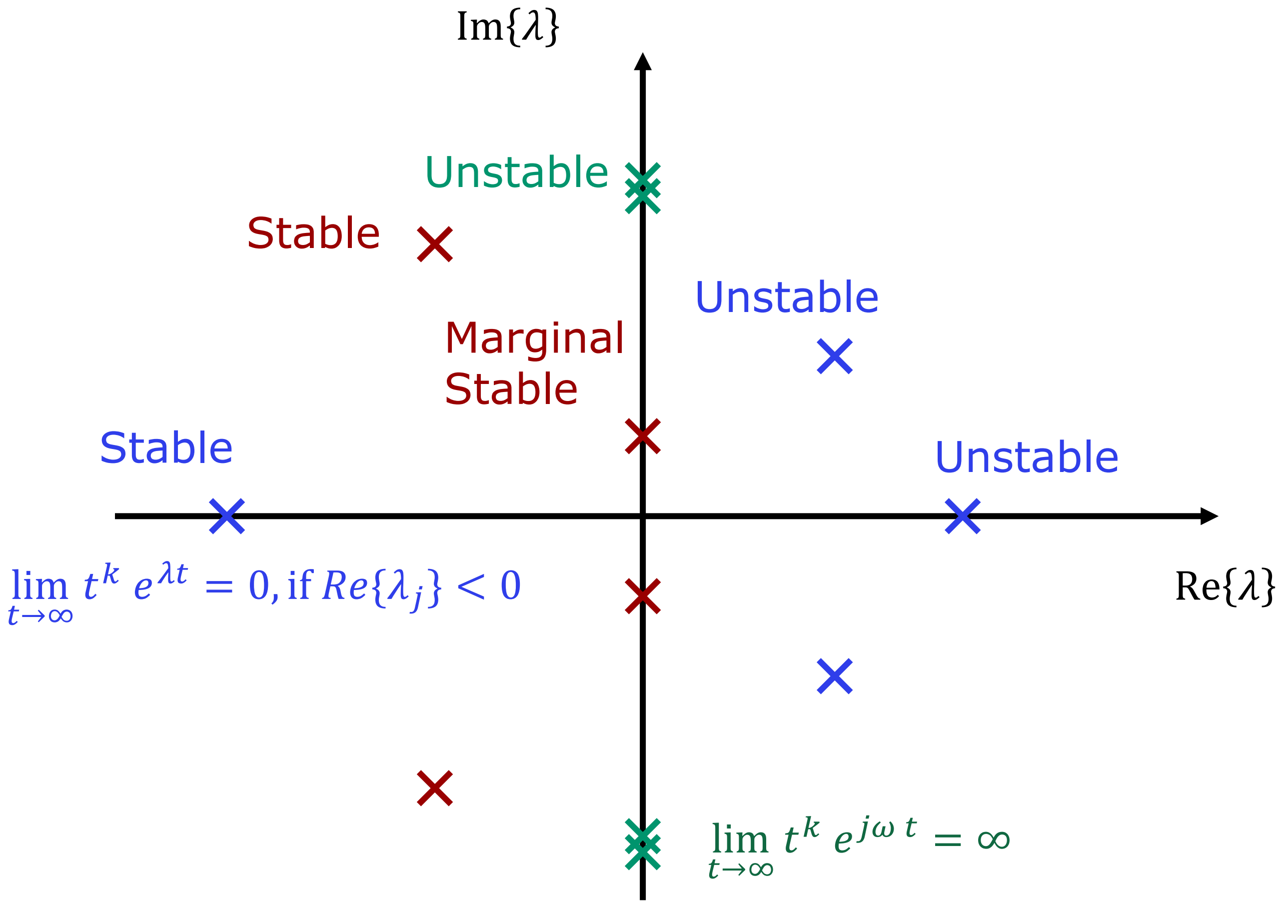
For repeated roots:

$$\lim_{t\to\infty} t^k\, e^{\lambda t} = 0, \qquad \text{if } Re\{\lambda_j\} < 0$$

$$\lim_{t\to\infty} t^k\, e^{j\omega\, t} = \infty, \qquad \text{if } Re\{\lambda_j\} = 0$$

For systems with repeated roots, non of these can be on the imaginary axis.

An unconditionally stable system has all its roots in the: LEFT HALF PLANE



Bounded Input- Bounded Output (BIBO) stability

If we do not know the natural modes of the system, we need an alternative means of determining if the system is stable.

In an experiment, we can provide an input and measure the output. We may also set up some relations to give us insight in what to expect of a stable system.

$$y_{zs}(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

$$|\,y_{zs}(t)| \leq \int_{-\infty}^{\infty} |h(\tau)||x(t-\tau)|d\tau$$

Bounded input:

$$|x(t-\tau)| \leq K_1$$

$$|\,y_{zs}(t)| \leq K_1 \int_{-\infty}^{\infty} |h(\tau)|d\tau$$

For an asymptotically stable system :

$$Re\{\lambda_j\} < 0$$

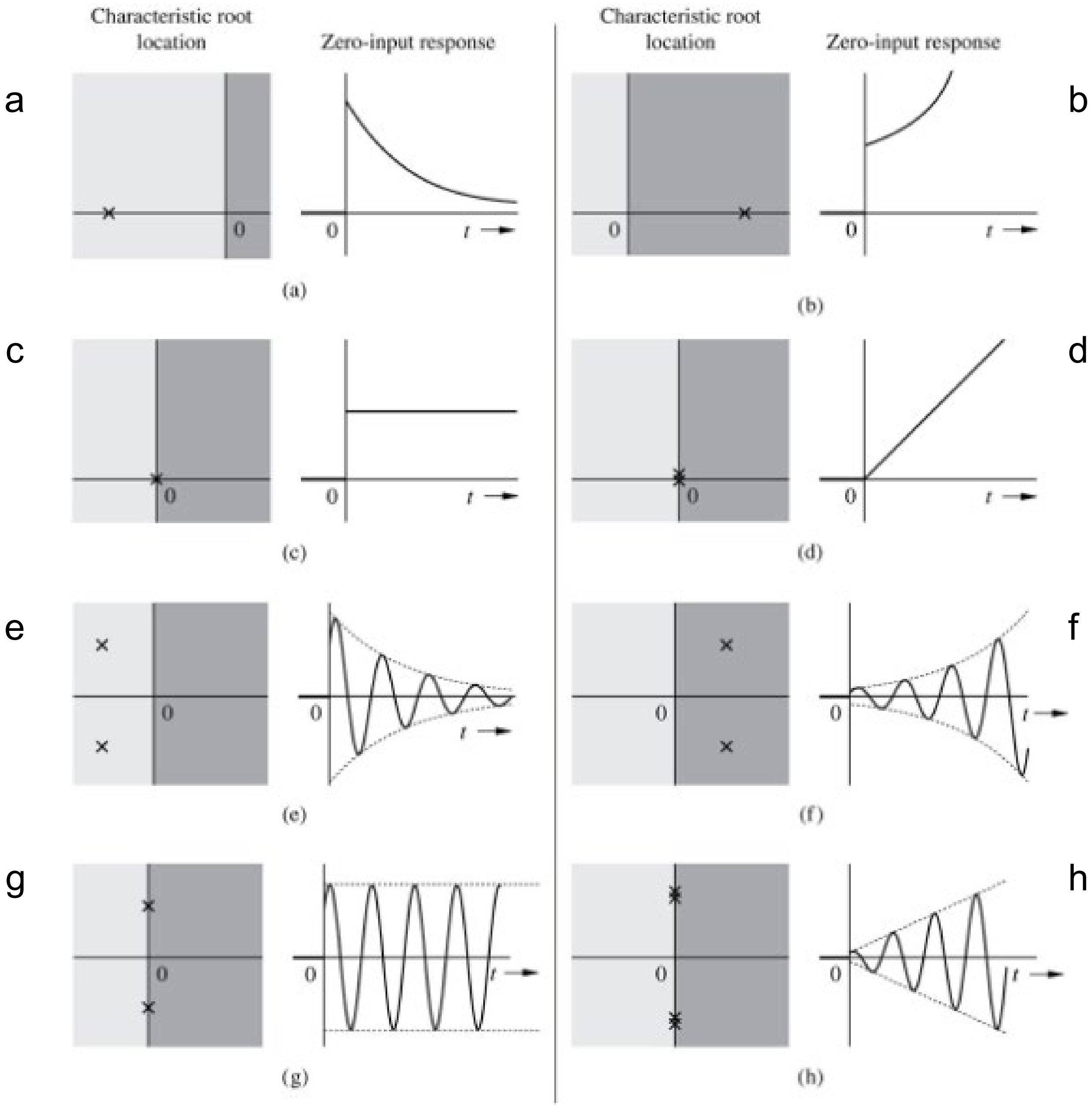
$$\int_{-\infty}^{\infty} |h(\tau)|d\tau \leq K_2$$

Bounded output

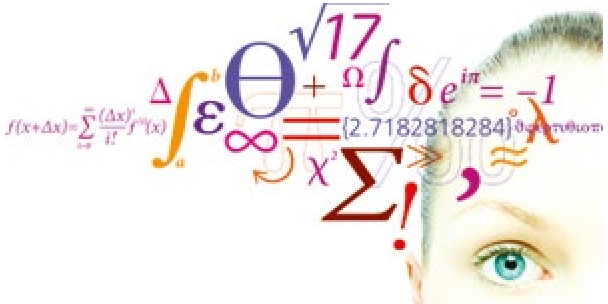
$$|\,y_{zs}(t)| \leq K_1 K_2$$

- An asymptotically stable system is always BIBO stable.
- BIBO stability ensures that the zero-state response is always amplitude bounded.
- BIBO stability only concerns what can be observed on the output (external).

Quick quiz



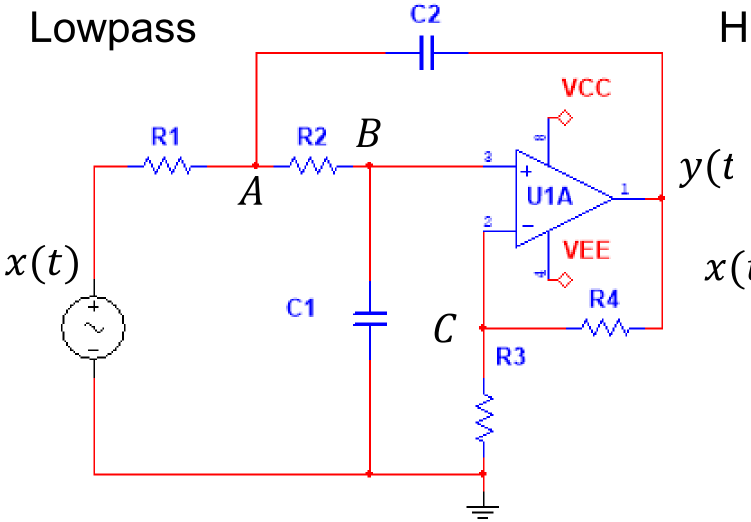
Characteristic root location
Zero-input response
Characteristic root location
Zero-input response
a
b
c
d
e
f
g
h
(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)



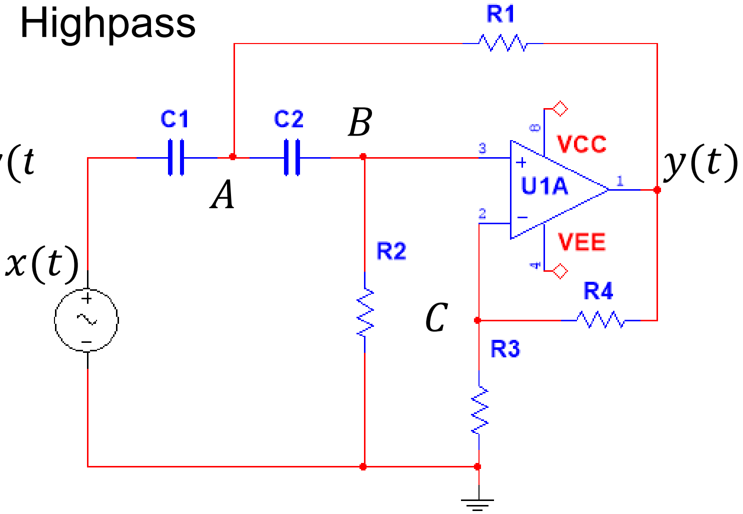
Problems

Sallen-Key filters used in this course

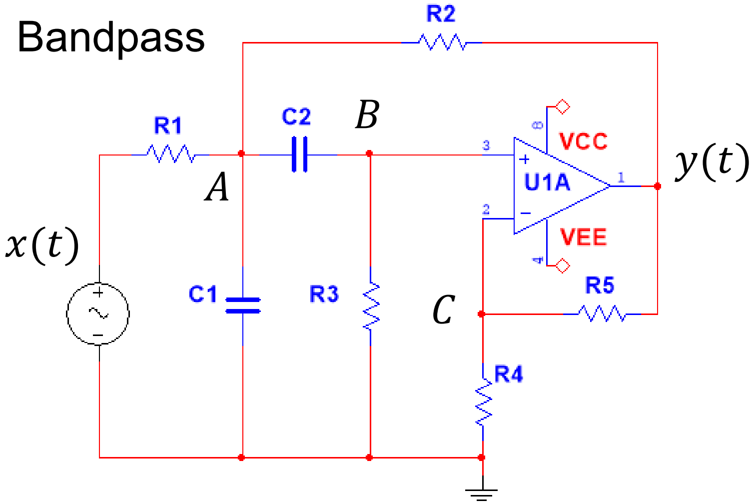
Lowpass



Highpass



Bandpass



Filter table

		Lowpass	Highpass	Bandpass
R_1	$k\Omega$	3.9894	1784.1	56.419
R_2	$k\Omega$	0.8865	892.06	34.117
R_3	$k\Omega$	1.0	1.0	149.72
R_4	$k\Omega$	1.0	1.0	1.0
R_5	$k\Omega$	–	–	1.0
C_1	nF	1795.2	1784.1	56.419
C_2	nF	398.94	3568.2	56.419
a_1		$2.83 \cdot 10^3$	$6.283 \cdot 10^{-1}$	$3.141 \cdot 10^1$
a_0		$3.95 \cdot 10^5$	$9.87 \cdot 10^{-2}$	$9.87 \cdot 10^4$
b_2		0	2	0
b_1		0	0	$6.283 \cdot 10^2$
b_0		$7.89 \cdot 10^5$	0	0

Filter 7: Lowpass filter - Convolution

To solve this problem, you will need the solutions to problems solved in weeks 1 and 2.

$$h(t) = 311.8(e^{-147.31\,t} - e^{-2680.2\,t})u(t)$$

1. Determine if the lowpass filter is asymptotically stable, marginally stable or unstable.
2. Plot $|H(j\omega)|$ in decibels.
3. Using Maple, apply the convolution integral to calculate the zero-state response to the input $x(t) = 2u(t)$.
4. Use the Maple package *DynamicSystems* to verify the result in (3).
5. As another validation, use KiCad/Spice to plot the step response.

Short answer to (3): $y_{zs}(t) = (4 + 0.23\,e^{-2680.2\,t} - 4.23\,e^{-147.25\,t})u(t)$

Filter 8: Highpass filter - Convolution

To solve this problem, you will need the solutions to problems solved in weeks 1 and 2.

$$h(t) = 2\delta(t) - 0.6283(2 - 031415\ t\ )e^{-031415\ t}u(t)$$

1. Determine if the highpass filter is asymptotically stable, marginally stable or unstable.
2. Plot $|H(j\omega)|$ in decibels.
3. Using Maple, apply the convolution integral to calculate the zero-state response to the input $x(t) = 2u(t)$.
4. Use the Maple package *DynamicSystems* to verify the result in (3).
5. As another validation, use KiCad/Spice to plot the step response.

Short answer to (3): $y_{zs}(t) = 4(1 - 0.31415\ t)e^{-0.31415\ t}u(t)$

Filter 9: Bandpass filter - Convolution

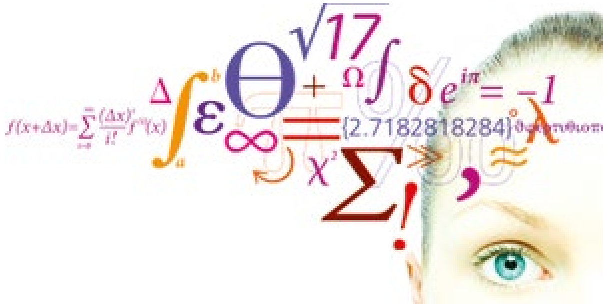
To solve this problem, you will need the solutions to problems solved in weeks 1 and 2.

$$h(t) = 628.3\ \mathrm{e}^{-15.7\,t}\cos(314\,t)u(t) - 31.4e^{-15.7\,t}\sin(314\,t)\,u(t)$$

1. Determine if the bandpass filter is asymptotically stable, marginally stable or unstable.
2. Plot $|H(j\omega)|$ in decibels.
3. Using Maple, apply the convolution integral to calculate the zero-state response to the input $x(t) = 2u(t)$.
4. Use the Maple package *DynamicSystems* to verify the result in (3).
5. As another validation, use KiCad/Spice to plot the step response.

Short answer to (3): $y_{zs}(t) = 4e^{-15.7\,t}\sin(313.76\,t)\,u(t) = 4e^{-15.7\,t}\cos\left(313.76\,t + \frac{\pi}{2}\right)u(t)$

Problem set (solutions)



Learning priorities from Problem Solving:

1. Why this method?
2. When is this method valid?
3. How is this method used?
4. What is the result?
5. How do I validate the result?

Solutions

Filter 7: Lowpass filter – Convolution (sol)

To solve this problem, you will need the solutions to problems solved in weeks 1 and 2.

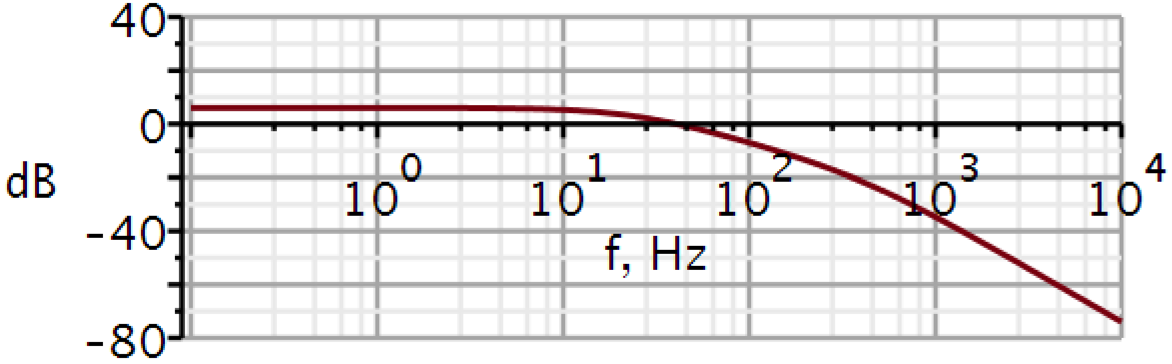
$$h(t) = 311.8(e^{-147.31\,t} - e^{-2680.2\,t})u(t)$$

1. Determine if the lowpass filter is asymptotically stable, marginally stable or unstable.
2. Plot $|H(j\omega)|$ in decibels.
3. Using Maple, apply the convolution integral to calculate the zero-state response to the input $x(t) = 2u(t)$.
4. Use the Maple package *DynamicSystems* to verify the result in (3).
5. As another validation, use KiCad/Spice to plot the step response.

Short answer to (3): $y_{zs}(t) = (4 + 0.23\,e^{-2680.2\,t} - 4.23\,e^{-147.25\,t})u(t)$

Filter 7: Lowpass filter – Convolution (sol)

1. The poles are all in the left-side plane. Hence the filter is asymptotically stable.



2. Amplitude characteristic:

Transfer function

$$H := \omega \rightarrow \frac{b2\cdot(\mathrm{j}\cdot\omega)^2 + b1\cdot(\mathrm{j}\cdot\omega) + b0}{(\mathrm{j}\cdot\omega)^2 + a1\cdot(\mathrm{j}\cdot\omega) + a0}: \qquad \# \textit{ 2nd order transfer function}$$

simplify(*evalf*(*H*(ω)))

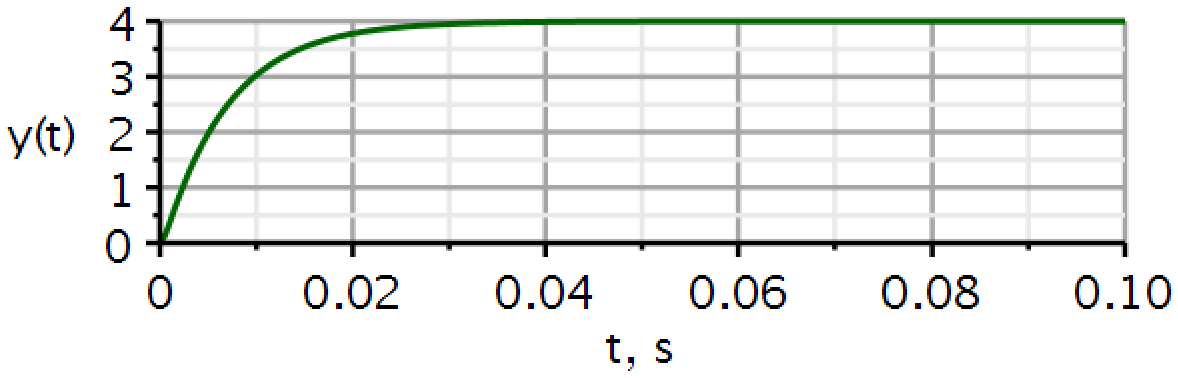
$$\frac{7.896292901\ 10^9}{-10000.\ \omega^2 + 2.827537902\ 10^7\ \mathrm{I}\ \omega + 3.948146451\ 10^9}$$

$$dB := \omega \rightarrow 20\cdot\log10(|H(\omega)|): \qquad \#\ \text{converts to decibel}$$

semilogplot(*dB*(2·π·*f*), *f* = 0.1 ..1E4, −80 ..40, *thickness* = 3, *axesfont* = ["Helvetica", "roman", 18], *axis*[2] = [*thickness* = 2.5], *axis*[1] = [*thickness* = 2.5], *labels* = ["f, Hz" , "dB"], *labelfont* = ["HELVETICA", 18], *numpoints* = 100, *gridlines*, *size* = [600, 200])

Filter 7: Lowpass filter – Convolution (sol)

3. Zero-state response using convolution:



$x1 := t \rightarrow 2 \cdot v(t)$:   # *input signal*

$h1 := t \rightarrow \left(311.75\, \mathrm{e}^{-147.31\, t} - 311.75\, \mathrm{e}^{-2680.2\, t}\right) \mathrm{Heaviside}(t)$:   # *impulse response*

$$evalf\left(\int_0^t x1(\tau) \cdot h1(t-\tau)\, \mathrm{d}\tau, 5\right)$$

$$3.9999 - 4.2326\, \mathrm{e}^{-147.31\, t} + 0.23263\, \mathrm{e}^{-2680.2\, t}$$

Short answer to (3): $y_{zs}(t) = (4 + 0.23\, e^{-2680.2\, t} - 4.23\, e^{-147.25\, t})u(t)$

Filter 7: Lowpass filter – Convolution (sol)

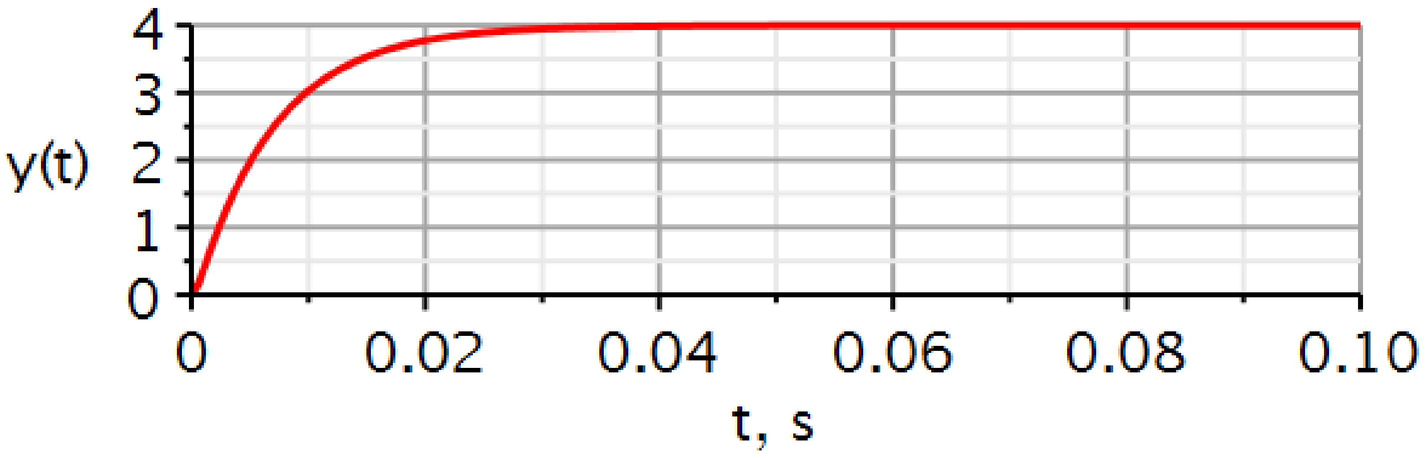
4. Use the Maple package *DynamicSystems* to verify the result.

$$sysLP := Coefficients\left(\frac{b0}{s^2 + a1 \cdot s + a0}\right):$$

$$vin := Step(2, 0, 0)$$

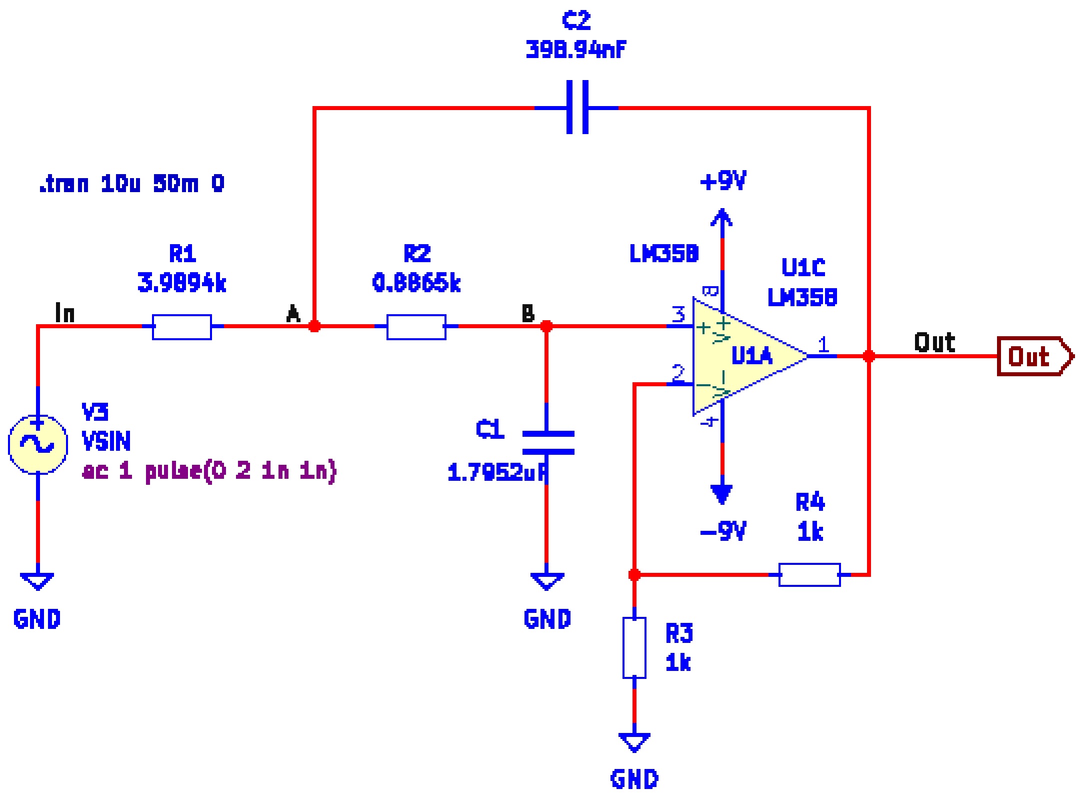
$$vin := 2\,\mathrm{Heaviside}(t)$$

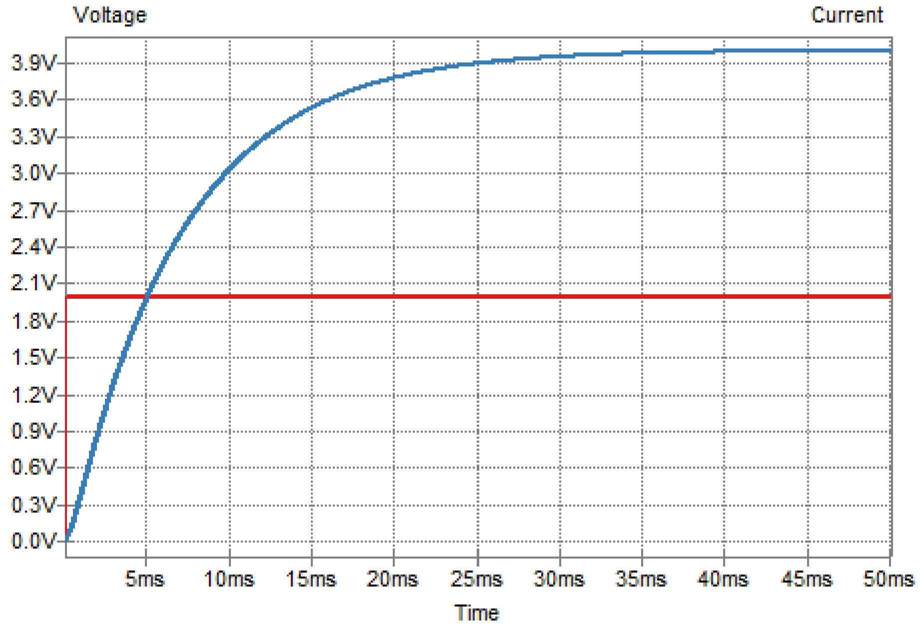
$ResponsePlot(sysLP, vin, duration = 0.1, color = [red], thickness = 3, axesfont = [\text{"Helvetica"}, \text{"ROMAN"}, 18], axis[2] = [thickness = 2.5], axis[1] = [mode = linear, thickness = 2.5], labels = [\text{"t, s"}, \text{"y(t)"}], labelfont = [\text{"HELVETICA"}, 18], gridlines, size = [600, 200])$



Filter 7: Lowpass filter – Convolution (sol)

5. As another validation, use KiCad/Spice to plot the step response.





Filter 8: Highpass filter – Convolution (sol)

To solve this problem, you will need the solutions to problems solved in weeks 1 and 2.

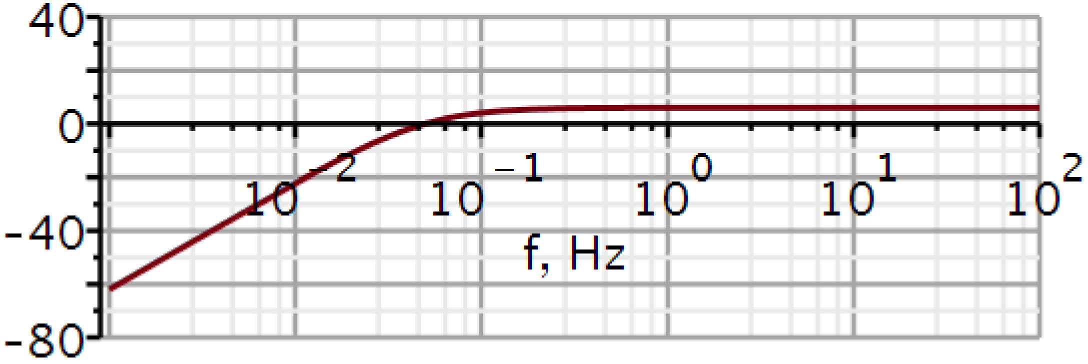
$$h(t) = 2\delta(t) - 0.6283(2 - 031415\ t\ )e^{-031415\ t}u(t)$$

1. Determine if the highpass filter is asymptotically stable, marginally stable or unstable.
2. Plot $|H(j\omega)|$ in decibels.
3. Using Maple, apply the convolution integral to calculate the zero-state response to the input $x(t) = 2u(t)$.
4. Use the Maple package *DynamicSystems* to verify the result in (3).
5. As another validation, use KiCad/Spice to plot the step response.

Short answer to (3): $y_{zs}(t) = 4(1 - 0.31415\ t)e^{-0.31415\ t}u(t)$

Filter 8: Highpass filter – Convolution (sol)

1. The poles are all in the left-side plane. Hence the filter is asymptotically stable.
2. Amplitude characteristic:



Transfer function

$$H := \omega \rightarrow \frac{b2\cdot(j\cdot\omega)^2 + b1\cdot(j\cdot\omega) + b0}{(j\cdot\omega)^2 + a1\cdot(j\cdot\omega) + a0}:\qquad \text{\# 2nd order transfer function}$$

$simplify(evalf(H(\omega)))$

$$-\frac{2.\,\omega^2}{-\omega^2 + 0.6283249520\,I\,\omega + 0.09870027409}$$

$$dB := \omega \rightarrow 20\cdot\log10(|H(\omega)|):\qquad \text{\# converts to decibel}$$

$semilogplot(dB(2\cdot\pi\cdot f), f = 0.001 ..1E2, -80 ..40, thickness = 3, axesfont = ["Helvetica", "roman", 18], axis[2] = [thickness = 2.5], axis[1] = [thickness = 2.5], labels = ["f, Hz" , "dB"], labelfont = ["HELVETICA", 18], numpoints = 100, gridlines, size = [600, 200])$

3. Zero-state response using convolution:

$x1 := t \rightarrow 2 \cdot v(t)$: # *input signal*

$h1 := t \rightarrow \left(0.19739\, e^{-0.31416\, t}\, t - 1.2566\, e^{-0.31416\, t}\right) \mathrm{Heaviside}(t)$: # *impulse response*

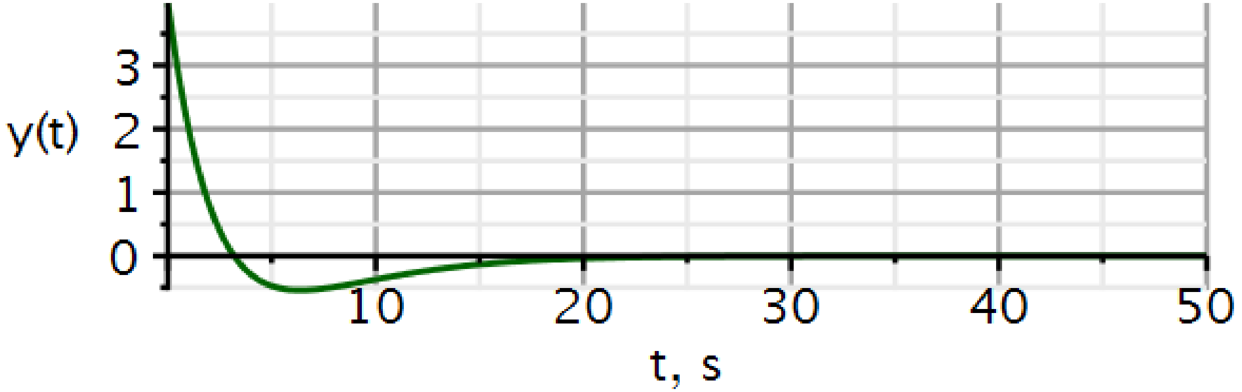
$$\boxed{2 \cdot x1(t)} + evalf\left(\int_0^t x1(\tau) \cdot h1(t - \tau)\, d\tau, 5\right)$$

$$4\, \mathrm{Heaviside}(t) - 3.9998 - 1.2566\, e^{-0.31416\, t}\, t + 3.9998\, e^{-0.31416\, t}$$

$y1 := t \rightarrow 2 \cdot x1(t) + conv(t \rightarrow x1(t), t \rightarrow h1(t))(t)$ # *output signal*

$$y1 := t \mapsto 2 \cdot x1(t) + conv(t \mapsto x1(t), t \mapsto h1(t))(t)$$

plot({*y1*(*t*)}, *t* = 0 ..50, *thickness* = 3, *color* = "DarkGreen", *axesfont* = ["Helvetica", "ROMAN", 18], *axis*[2] = [*thickness* = 2.5], *axis*[1] = [*mode* = *linear*, *thickness* = 2.5], *labels* = ["t, s" , "y(t)"], *labelfont* = ["HELVETICA", 18], *gridlines*, *size* = [600, 200])



When $b_n \neq 0$, remember to add $b_n \delta(t)$ to the impulse response.

Short answer to (3): $y_{zs}(t) = 4(1 - 0.31415\, t)e^{-0.31415\, t} u(t)$

4. Use the Maple package *DynamicSystems* to verify the result.

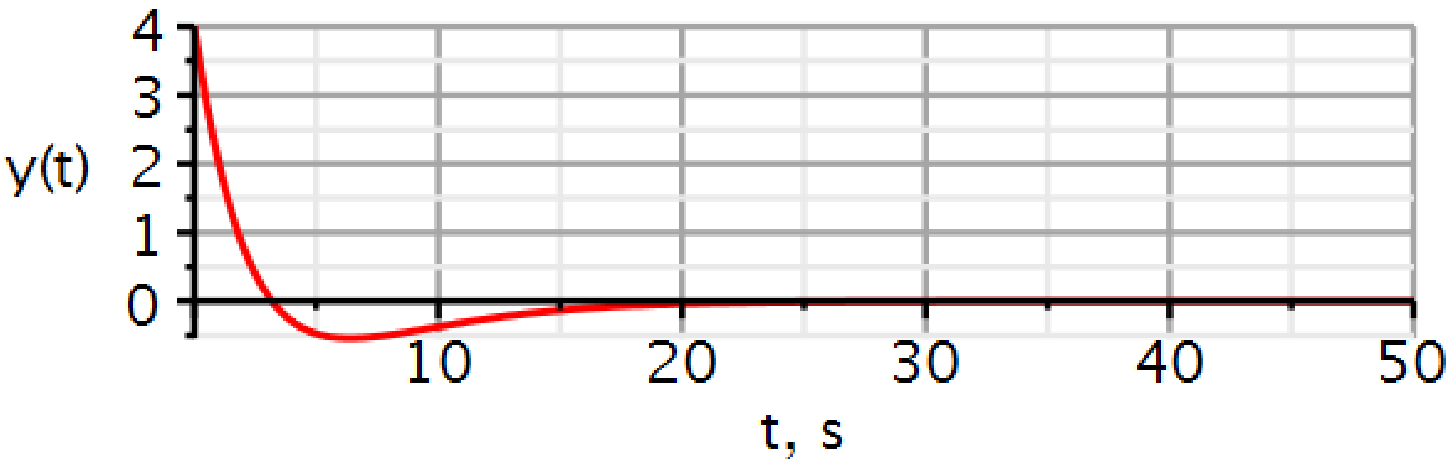
Dynamic systems

$$sysHP := Coefficients\left(\frac{b2 \cdot s^2}{s^2 + a1 \cdot s + a0}\right):$$

$vin := Step(2, 0, 0)$

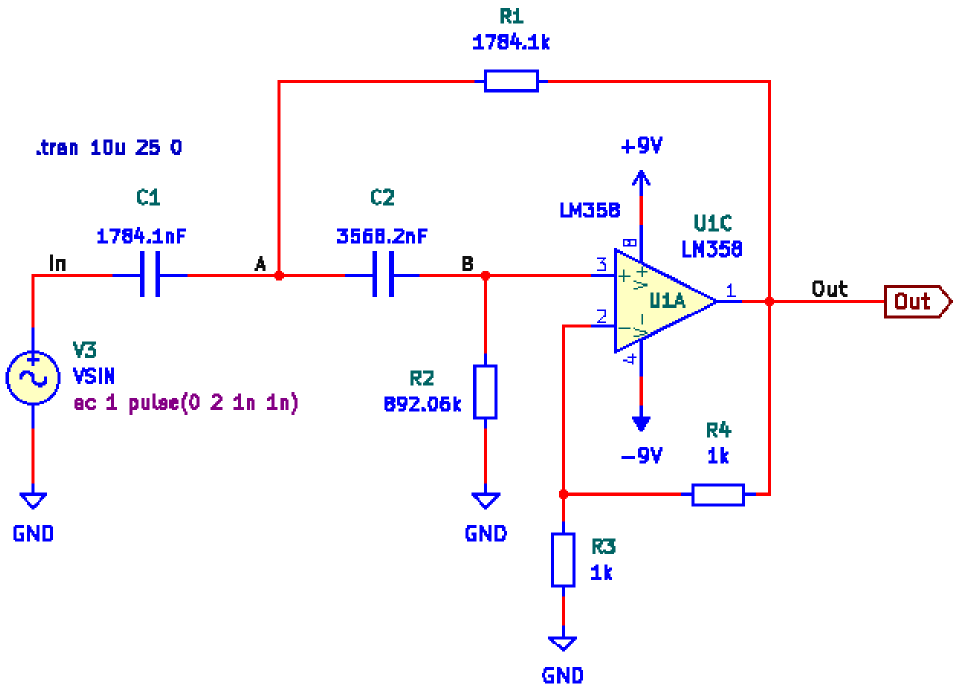
$$vin := 2\,\mathrm{Heaviside}(t)$$

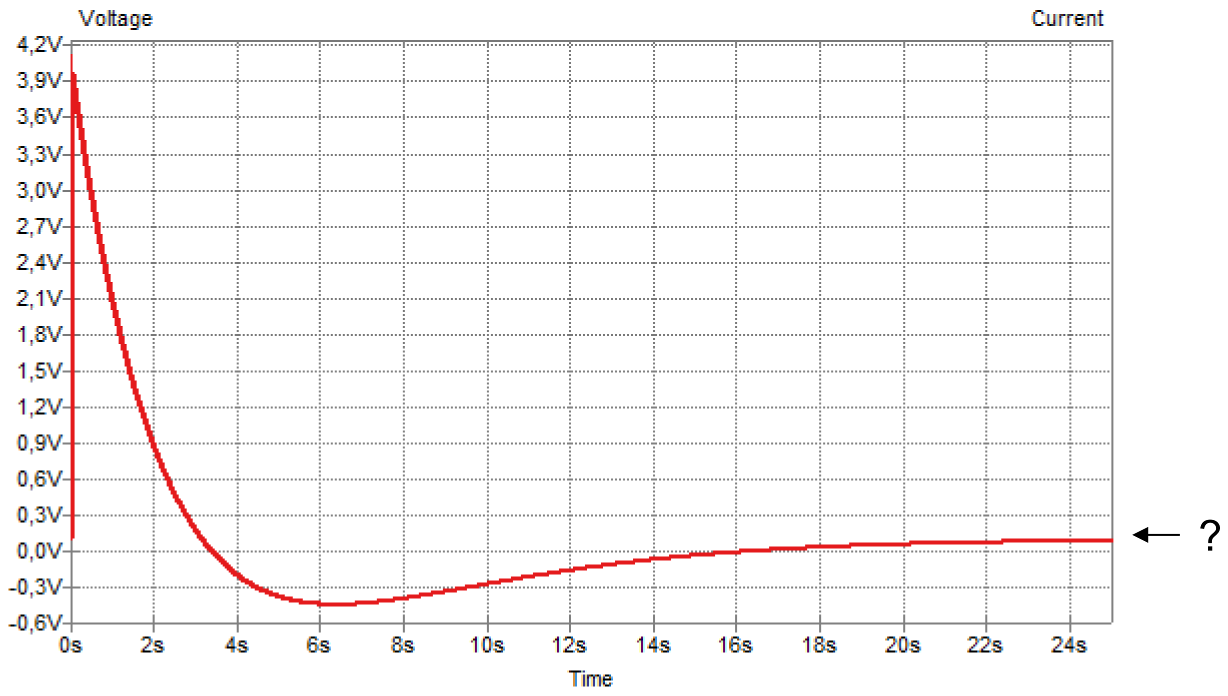
$ResponsePlot(sysHP, vin, duration = 50, color = [red], thickness = 3, axesfont = [\text{"Helvetica"}, \text{"ROMAN"}, 18], axis[2] = [thickness = 2.5], axis[1] = [mode = linear, thickness = 2.5], labels = [\text{"t, s"}, \text{"y(t)"}], labelfont = [\text{"HELVETICA"}, 18], gridlines, size = [600, 200])$



Filter 8: Highpass filter – Convolution (sol)

5. As another validation, use KiCad/Spice to plot the step response.





Filter 9: Bandpass filter – Convolution (sol)

To solve this problem, you will need the solutions to problems solved in weeks 1 and 2.

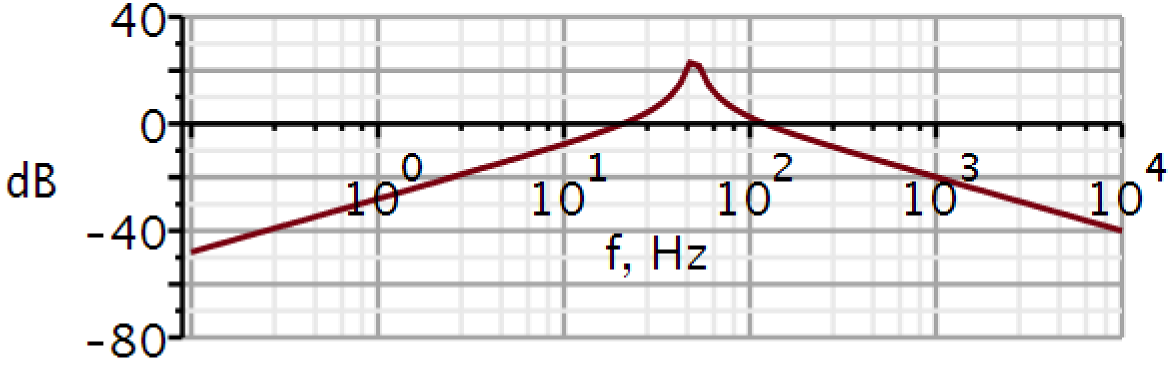
$$h(t) = 628.3\ \mathrm{e}^{-15.7\,t}\cos(314\,t)u(t) - 31.4e^{-15.7\,t}\sin(314\,t)\,u(t)$$

1. Determine if the bandpass filter is asymptotically stable, marginally stable or unstable.
2. Plot $|H(j\omega)|$ in decibels.
3. Using Maple, apply the convolution integral to calculate the zero-state response to the input $x(t) = 2u(t)$.
4. Use the Maple package *DynamicSystems* to verify the result in (3).
5. As another validation, use KiCad/Spice to plot the step response.

Short answer to (3): $y_{zs}(t) = 4e^{-15.7\,t}\sin(313.76\,t)\,u(t) = 4e^{-15.7\,t}\cos\left(313.76\,t + \frac{\pi}{2}\right)u(t)$

Filter 9: Bandpass filter – Convolution (sol)

1. The poles are all in the left-side plane. Hence the filter is asymptotically stable.
2. Amplitude characteristic:



Transfer function

$$H := \omega \to \frac{b2\cdot(j\cdot\omega)^2 + b1\cdot(j\cdot\omega) + b0}{(j\cdot\omega)^2 + a1\cdot(j\cdot\omega) + a0} : \qquad \text{\# 2nd order transfer function}$$

simplify(*evalf*($H(\omega)$))

$$\frac{1.96349251\ 10^{8}\ I\ \omega}{-312500.\ \omega^2 + 9.814365906\ 10^{6}\ I\ \omega + 3.084214098\ 10^{10}}$$

$$dB := \omega \to 20\cdot \log10(|H(\omega)|) : \qquad \text{\# converts to decibel}$$

semilogplot($dB(2\cdot\pi\cdot f)$, $f = 0.1\,..1E4$, $-80\,..40$, *thickness* = 3, *axesfont* = ["Helvetica", "roman", 18], *axis*[2] = [*thickness* = 2.5], *axis*[1] = [*thickness* = 2.5], *labels* = ["f, Hz" , "dB"], *labelfont* = ["HELVETICA", 18], *numpoints* = 100, *gridlines*, *size* = [600, 200])

Filter 9: Bandpass filter – Convolution (sol)

3. Zero-state response using convolution:

$x1 := t \to 2 \cdot v(t):$ # *input signal*

$h1 := t \to \left(-31.447\, e^{-15.703\, t} \sin(313.76\, t) + 628.33\, e^{-15.703\, t} \cos(313.76\, t)\right) \mathrm{Heaviside}(t):$ # *impulse response*

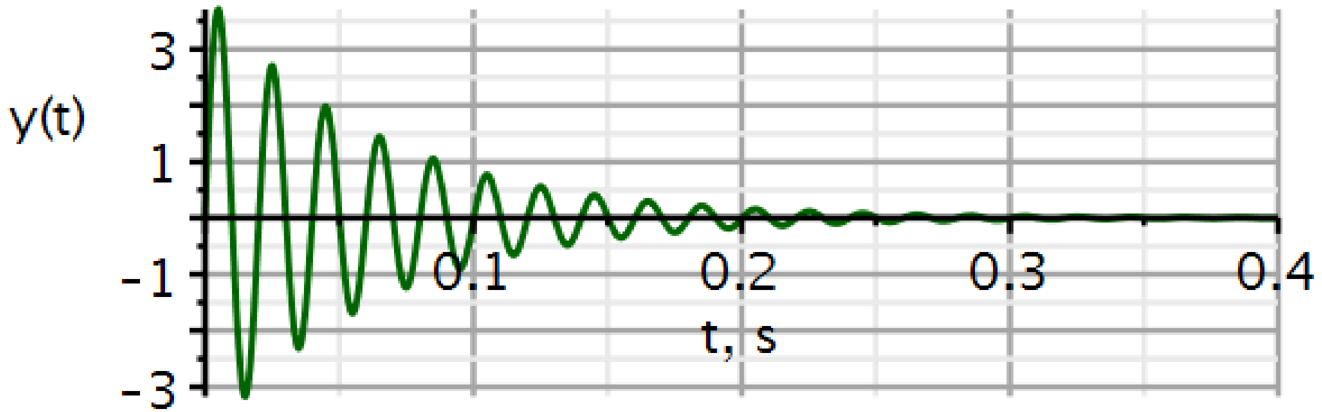
$$evalf\left(\int_0^t x1(\tau) \cdot h1(t-\tau)\, d\tau, 5\right)$$

$$-2.9330\ 10^{-6} + 2.9330\ 10^{-6}\, e^{-15.703\, t} \cos(313.76\, t) + 4.0052\, e^{-15.703\, t} \sin(313.76\, t)$$

$y1 := t \to conv(t \to x1(t), t \to h1(t))(t)$ # *output signal*

$y1 := t \mapsto conv(t \mapsto x1(t), t \mapsto h1(t))(t)$

plot({ *y1*(*t*) }, *t* = 0 ..0.4, *thickness* = 3, *color* = "DarkGreen", *axesfont* = ["Helvetica", "ROMAN", 18], *axis*[2] = [*thickness* = 2.5], *axis*[1] = [*mode* = *linear*, *thickness* = 2.5], *labels* = ["t, s" , "y(t)"], *labelfont* = ["HELVETICA", 18], *gridlines*, *size* = [600, 200])



Short answer to (3): $y_{zs}(t) = 4e^{-15.7\, t} \sin(313.76\, t)\, u(t) = 4e^{-15.7\, t} \cos\left(313.76\, t + \frac{\pi}{2}\right) u(t)$

4. Use the Maple package *DynamicSystems* to verify the result.

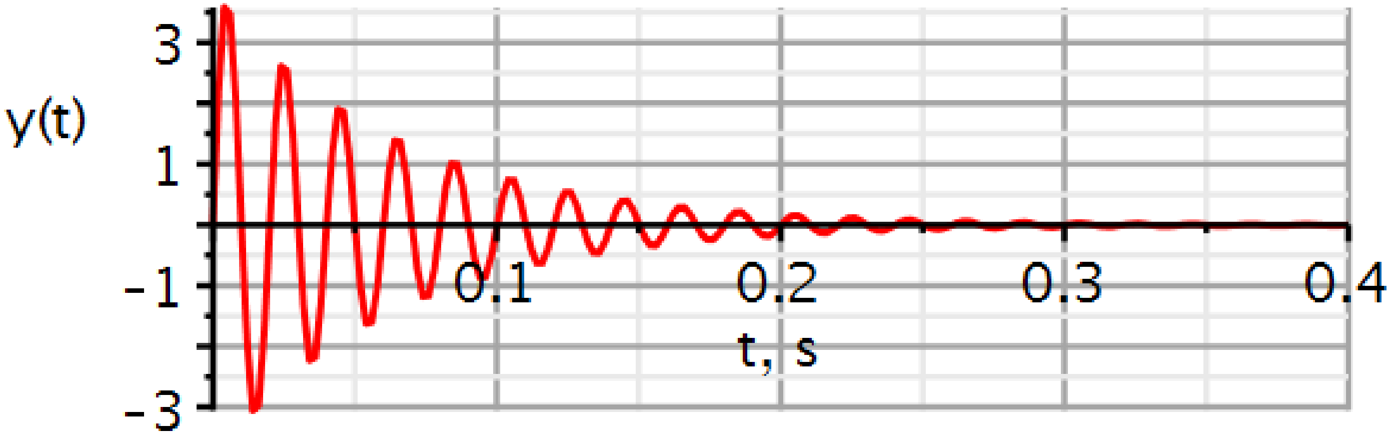
Dynamic systems

$$sysBP := Coefficients\left(\frac{b1 \cdot s}{s^2 + a1 \cdot s + a0}\right):$$

$vin := Step(2, 0, 0)$

$$vin := 2\,\mathrm{Heaviside}(t)$$

ResponsePlot(*sysBP*, *vin*, *duration* = 0.4, *color* = [*red*], *thickness* = 3, *axesfont* = ["Helvetica", "ROMAN", 18], *axis*[2] = [*thickness* = 2.5], *axis*[1] = [*mode* = *linear*, *thickness* = 2.5], *labels* = ["t, s" , "y(t)"], *labelfont* = ["HELVETICA", 18], *gridlines*, *size* = [600, 200])



Filter 9: Bandpass filter – Convolution (sol)

5. As another validation, use KiCad/Spice to plot the step response.